



Virtual try-on for sustainable fashion retail: mitigating return waste through consumer decision accuracy under privacy concerns

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ABSTRACT

Virtual Try-On (VTO) technology has emerged as a promising solution for reducing high product return rates in online fashion retail, which contribute significantly to environmental waste. This study examines how VTO improves product selection accuracy and reduces return-related waste while considering the influence of consumer data privacy concerns. Drawing on the Technology Acceptance Model and Privacy Calculus Theory, the study adopts a mixed-methods approach involving survey data from 402 online fashion shoppers in India and semi-structured interviews for deeper consumer insights. The findings reveal that VTO significantly enhances consumers' confidence in selecting suitable products, leading to fewer product returns and lower return-related waste. Product selection accuracy further mediates the relationship between VTO usage and returns reduction. However, concerns regarding personal data collection and privacy reduce consumers' willingness to fully engage with VTO technologies. The qualitative findings additionally show that consumers value VTO for minimizing uncertainty related to size and fit, although apprehensions about biometric and personal data sharing remain important barriers. The study contributes theoretically by integrating technology adoption, privacy concerns, and sustainability outcomes within the context of digital fashion retailing. Practically, it highlights the need for online retailers to develop transparent, secure, and user-friendly VTO systems that enhance consumer trust while supporting sustainable e-commerce practices.

1. Introduction

The rapid growth of online fashion retailing has significantly increased product return rates, creating major environmental and operational challenges for retailers. Shopping on the internet has gained popularity among the consumers due to its convenience, diversity and time-saving. The online sales have particularly been rising in the fashion and lifestyle sectors. Nevertheless, among the most significant issues that e-commerce platforms have experienced is product returns which are very high especially in the apparel and fashion category [1–6]. Studies indicate that product returns in online apparel shopping are primarily driven by uncertainty related to size, fit, colour, and product appearance, as consumers are unable to physically evaluate products before purchase [7,8]. Studies further show that ill-fitting products account for a substantial proportion of apparel returns in e-commerce,

resulting in increased reverse logistics costs, packaging waste, and carbon emissions associated with transportation and reprocessing [8,9]. Such high rates of returns contribute to higher logistics expenses and severe environmental issues like the over-packaging of products, an increase in transportation emissions, and inefficient reverse logistics [10–13]. Consequently, fashion retailers are increasingly adopting digital technologies such as Augmented Reality (AR) and Virtual Try-On (VTO) systems to improve purchase confidence and minimise return-related waste [14].

VTO technologies enable consumers to digitally visualise apparel, cosmetics, eyewear, or accessories on their own image or avatar before making purchase decisions. These technologies are increasingly being adopted by major fashion and beauty retailers to improve customer experience and decision accuracy. For instance, platforms such as Myntra, Ajio, Nykaa Fashion, Amazon Fashion, and Lenskart have

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introduced AR-based try-on features that allow customers to assess fit, appearance, and style virtually before purchase [7]. Internationally, collaborations such as Nike × Snapchat have created immersive AR-based shopping experiences, while retailers are increasingly integrating AI-powered virtual fitting technologies to enhance consumer engagement and sustainability outcomes [7]. Similarly, technology providers such as Zeekit, Vue.ai, Wanna, Fits.Me, and True Fit support fashion retailers in implementing personalised virtual fitting solutions that improve shopping confidence and reduce uncertainty during online purchasing [8].

Online retailers have been progressively using VTO technology in order to overcome these problems. VTO is a digital product trial featuring clothes, shoes, accessories, cosmetics and so on; a consumer can try fashion products virtually using AR, artificial intelligence (AI), and computer vision technologies [15–17]. Most recent developments in computer vision have made possible image-based VTO systems which do not require paired training data, making the online fashion visualization more flexible and realistic [18]. Studies indicate that VTO enhances the visualisation skill of the consumers and the confidence they have when purchasing a product thus eliminates the factors of uncertainty in online shopping [19–21]. VTO is also considered a green retail innovation because it can help in achieving sustainability through the reduction of waste, emission and resource use due to its potential of reducing unnecessary returns [10,11,22]. VTO and VR (Virtual Reality) shopping technologies enhance the accuracy of product selection by offering realistic previews and offering customized advice, as well as direct consumer focus to essential product information [23]. Recent studies suggest that VTO technologies reduce uncertainty in online shopping by improving consumers' confidence in product selection and minimising the likelihood of ordering multiple sizes or returning unsuitable products [9]. Beyond improving shopping convenience, these technologies also contribute to sustainable e-commerce practices by reducing unnecessary transportation, packaging waste, and reverse logistics activities associated with returns [8,24].

Although these are the advantages, there is a collection and processing of personal and biometric data in the form of facial images, body measurements and appearance related information in the VTO technology. This creates serious issues of privacy of data, security of data and ethical application of consumer information [20,25–28]. Research indicates that the privacy issue can adversely influence the willingness of the consumers to use modern digital technologies even when there are obvious advantages to the technology [26,29]. This presents a dilemma in terms of e-commerce sites to strike the right balance between innovation and consumer trust.

The general issue that the current study will achieve is the absence of a comprehensive knowledge of whether the VTO technology could efficiently be used to meet the green retail objectives and at the same time safeguard the privacy of consumer data. Although VTO is marketed as a measure to lower the rate of returns and enhance the shopping experience, still, high rates of returns are considered the significant problem of online stores [1,30]. Simultaneously, there is a growing awareness of the privacy threats and consumers are hesitant when it comes to providing personal information. Unless users are convinced of VTO systems, they might shun them hence limiting the efficiency of the technology [20,28]. Thus, the challenge lies in balancing sustainability benefits with ethical and transparent data practices.

The VTO technology has a number of obstacles that do not allow it to be used successfully. One of the major issues is that data handling practices are not transparent. Lots of privacy policies are difficult to comprehend, so it is difficult to tell what happens with the information of the customers. The result of this is the confusion and loss of trust [25]. Further, threats of data attacks and biometric misuse make users feel even more victimized [26,27]. Technological accuracy is another obstacle. In case VTO systems do not give realistic or accurate product representations, consumers might also be dissatisfied and revert [31, 32]. They are also not adopted because of high implementation costs and

technical requirements, particularly to small and medium-sized retailers.

This study is significant as it addresses the point of meeting the three areas of technology, sustainability, and consumer privacy that are one of the most critical in the contemporary e-commerce. There are products that are returned contributing to the degradation of the environment especially in the fashion industry [3,13]. VTO technologies provide a chance to minimize this influence by promoting the decision to purchase [33,34]. Meanwhile, consumer privacy should be preserved to gain trust and make digital innovations acceptable in the long term [35]. This research is of particular interest to the Indian context, in which online shopping is growing at an impressive rate, yet the knowledge about data protection and sustainable consumption is developing [1].

Current literature shows that the VTO technology has a positive effect on consumer purchase intention, interaction, and decision comfort [19,32,36,37]. Research has also pointed out the application of AR-based VTO in increasing the shopping experience and perceived risk [15,20,36]. Simultaneously, studies of the privacy paradox have indicated that despite a consumer placing a high value on his/her privacy, people can still use VTO systems because of perceived advantages [20, 35].

Nonetheless, the majority of the research looks at the technological performance, consumer behaviour, or privacy issues individually, and creates a discontinuous picture on the comprehensive effect of VTO. A number of research gaps have been found based on literature. To begin with, empirical studies that combine the use of VTO, the reduction of return waste, and the issues of data privacy in one research are scanty [28,33]. Secondly, no mixed-methods studies have integrated quantitative surveys with qualitative interviews to elicit behavioural patterns and perceptions of the consumer based on a deeper understanding [34, 38]. Third, there is insufficient research on emerging markets such as India, where consumer behaviour and regulatory conditions together with technology adoption are not similar to developed economies [1, 39]. Also, there is limited research examining the intention to relate VTO use with the sustainability performance among consumers [11,22].

This study is necessary because of the rising use of VTO technology and the emergence of more concerns about environmental sustainability and safety of data privacy. Although VTO is placed as a green retail solution, the success would rely on consumer trust and acceptance. The manner in which the consumers view both the advantages of sustainability and the risks of privacy is the key to responsible technological use. The suggested study is expected to present findings that would assist online stores in enhancing the level of transparency, creating privacy-aware systems, and refraining from unfriendly online shopping patterns. The proposed research project adds to the existing literature by incorporating the idea of VTO technology, green retail purpose, and consumer data privacy into one research. It is a mixed-methodology approach, which incorporates both survey and interviewing data to give a comprehensive view of the consumer perceptions. The research also provides empirical data that is specific to India, which is one of the gaps in the current literature. More than that, it offers realistic application to the e-commerce platforms that aim to minimize the waste of returns and ensure the trust of consumers.

This study has a mixed-methods research design in terms of its methodology. The quantitative stage is a structured questionnaire to be held among those online shoppers who have experience in using VTO features. Relationships between VTO use and sustainability perception, reduction of returns and privacy issues are analyzed statistically. The qualitative stage will be represented by the semi-structured interviews investigating the consumer experience, trust concerns, and attitudes to VTO technology. Combination of the two methods will provide a thorough and balanced analysis of the research problem.

2. Literature review

This section reviews the existing literature related to VTO

technology, online shopping behaviour, product return reduction, sustainability outcomes, and data privacy concerns in digital fashion retailing. The section also establishes the theoretical foundation of the study by drawing upon the Technology Acceptance Model (TAM) developed by [40] and Privacy Calculus Theory proposed by [41]. Furthermore, the review synthesises technological, behavioural, and sustainability perspectives to explain how VTO adoption may improve product selection accuracy and reduce return-related waste while considering the moderating role of data privacy concerns. The review additionally identifies existing research gaps and develops the hypotheses for the proposed conceptual framework. The section combines the technological, behavioural, and sustainability views to describe how VTO implementation could have an impact on the accuracy of product selection and minimise returns-related waste, and takes into account the mediating influence of the issue of data privacy concerns.

2.1. Overview of existing theories and concepts

The skyrocketing development of e-commerce, especially in the retailing of fashion and apparel, has made consumers more dependent on digital technologies in their purchasing choices. Online shopping, however, does not involve touching and feeling the product and hence there is a lot of uncertainty regarding the size of the product, its fit as well as its appearance, which leads to a high rate of returns [2–4,6,12]. As a solution to this problem, retailers have implemented VTO technologies, which enable consumers to see products virtually before buying them [21,33]. Perceived usefulness, realism, and ease of use influence consumer adoption of digital fashion products and AI-driven VTO systems [42,43]. The theoretical foundation of this study is primarily based on the Technology Acceptance Model (TAM) introduced by [40] and Privacy Calculus Theory developed by [41]. TAM explains how perceived usefulness and perceived ease of use influence users' acceptance and adoption of new technologies. In contrast, Privacy Calculus Theory explains how consumers evaluate the trade-off between perceived benefits and perceived privacy risks before deciding whether to adopt a digital technology. These theories are particularly relevant in the context of VTO technologies, where consumers simultaneously evaluate shopping convenience, decision accuracy, and privacy concerns. The Technology Acceptance Model (TAM) is among the most popular models that are used to explicate user acceptance of new technologies. TAM suggests that the attitudes and behavioural intention of the users towards using a technology are determined by two important beliefs, namely Perceived Usefulness (PU) and Perceived Ease of Use [44]. PU in the context of VTO is the degree to which consumers feel that the technology enhances their shopping performance by allowing them to select products more precisely, becoming more confident, and minimizing the risk of purchases [19,45]. Another construct based on TAM is Adoption Intention (AI), which indicates the willingness of the consumers to use VTO technology in online shopping. The recent works have extended TAM to analyze immersive and AI-based fueled retail technologies like AR, VR, and VTO applications [15,46]. According to such studies, consumers tend to be more willing to use and use such technologies as they consider them helpful and reliable in the decision-making process. The combination of chatbots, VTO, and online payment services facilitates customer satisfaction and overall interests in online shopping [47]. VTO experience in this study is conceptualised as a higher-order construct that would comprise of perceived usefulness and adoption intention that would capture cognitive evaluation as well as behavioural readiness. Having been developed alongside technology acceptance, data privacy has become a vital issue in online business. VTO technologies commonly demand the use of personal data like pictures, body measurements, or biometrics, which is an issue of data abuse and surveillance ([26,27,48]a; [20,28]).

According to Privacy Calculus Theory, consumers consider the perceived costs of a technology and the perceived privacy risks against the perceived benefits of a technology before they make a decision on

whether to adopt it or not [35]. When the benefits perceived surpass the perceived risks, users become more likely to adopt the technology; otherwise, they can avoid it or restrict their use. The issue of data privacy is negative and positive at the same time. Fear of data leakage, misuse, or transparency lack are negative data privacy concerns (DPCN), which can decrease the trust and use of VTO applications [27]. Conversely, positive data privacy perceptions (DPCP) do occur when the consumer believes that the platforms are using data responsibly, and this can encourage subsequent use and usage [25]. This paper addresses the issue of privacy as a moderator that may affect the effectiveness of VTO in the process of waste reduction at the return (see Table 1)

2.2. Review of empirical studies

Empirical studies that explore the importance of VTO and other similar technologies in influencing online consumer behaviour have a significant amount of data available. Preliminary research indicated that VTO applications increase user interaction, pleasure, and confidence in purchasing the product since they facilitate a more interactive buying process [31,32]. Several studies have focused on the technical effectiveness of VTO systems, highlighting improvements in visual realism, pose estimation, and garment fitting accuracy using AI and deep learning models [15,17,52]. The development of these technologies has improved the virtual evaluation of products by consumers and thus quality of the decisions made by consumers. Empirical data would indicate that proper visualisation leads to the increase of correct product choice and, in particular, the accuracy of the choices concerning the size and fit characteristics [11,15,16]. Online shopping has had a great amount of research on the subject of return behaviour in terms of operational and sustainability. The studies show that high returns result in high economic and environmental costs because of a reverse logistics system, repackaging, and wastes [3,10,53]. Examples in literature on fashion e-commerce indicate that size and fit discrepancy is among the most frequent factors that contribute to returns, with a huge portion of unnecessary waste being represented [5,13]. New empirical research has begun to associate the use of VTO with lower returns. For example, Hwangbo et al. [37] established that 3D VTO solutions are a significant way of increasing customer satisfaction and decreasing product returns. Similarly, Gao & Liang [19] stated that AI try-on products increase the comfort of the decision and decrease returns on impulsive buying. The results indicate that VTO may be directly applied in reducing a return waste through enhanced accuracy of the decisions made prior purchases.

Nevertheless, the importance of privacy issues as an adoption barrier is also supported by empirical evidence. Ragab et al. [28] determined privacy risks in various VTO applications that could deter users to embrace the technology to the latter. Youn et al. [20] noticed a privacy paradox, with the consumers enjoying the advantages of VTO but are still reluctant to adopt it because of data privacy concerns. Mutambik et al. [35] also substantiated that the issue of privacy adversely affects trust and the use of social commerce technologies.

In general, although the current research has sufficient evidence in favour of the advantages of VTO in terms of promoting purchase decisions and minimizing returns, not many studies may incorporate technological acceptance, privacy issues, and sustainability results into one study. This gap shows the necessity of a comprehensive approach that integrates TAM and Privacy Calculus Theory to explain how the introduction of VTO may result in the decrease in waste of returns with considering privacy-related obstacles [28,33,34].

2.3. Identification of key variables and constructs

This paper is based on a series of variables related to one another to answer the question of the contribution of VTO technology to the sustainable results in online shopping. VTO is the main independent variable since it reflects the total interaction and experience of the

Table 1

The list of existing studies.

Author(s) & Year	Context	Key Construct(s)	Method	Key Findings	Gap Identified
[40]	Technology adoption	Perceived Usefulness (PU), Perceived Ease of Use (PEOU)	Conceptual / Empirical validation	PU and PEOU determine behavioural intention toward technology use	Does not consider immersive retail tech like VTO or sustainability outcomes
[31]	Online fashion retail	Virtual Try-On (VTO)	Experimental study	VTO improves purchase confidence and product evaluation	No discussion on privacy or return waste
[37]	AR/Virtual fashion	Perceived usefulness	Experimental	Digital visualization reduces uncertainty in online apparel	No integration with return reduction behavior
[49]	Immersive retail technology	AR/VR adoption	Qualitative	Usefulness and reliability increase technology usage	Privacy concerns not incorporated
[35]	Privacy calculus in digital tech	Perceived benefits vs privacy risks	Quantitative	Adoption occurs when benefits outweigh risks	Not applied specifically to VTO in sustainable retail
[27]	Data privacy & e-commerce	Data Privacy Concerns (DPCN)	Review	Privacy fears reduce trust and technology adoption	Does not test moderating effect on sustainability outcomes
[25]	Responsible data usage	Positive Data Privacy Perception (DPCP)	Review	Responsible data handling increases user engagement	No link to return waste reduction
[50]	Indian e-commerce	Return rates in apparel	Secondary data analysis	High return rates due to size and fit issues	No technological solution analysis
[51]	Indian retail market	Online fashion growth	Industry report	Growing dependence on digital fashion retail	Lacks academic behavioural testing

Source: Compiled by the Author

consumers with VTO technology on the Internet. VTO and AR technology enhancement boost the consumer experience, accuracy in decision-making, and visualization of products by enabling the production of images with contextual considerations, the visualization of outfits, and user-friendly interaction design, which result in a lower rate of returns and sustainable fashion [15,54–56]. The construct of VTO Technology was conceptualised based on consumers' perceptions of usefulness, usability, and adoption intention toward VTO systems. The measurement items were adapted from studies examining VTO adoption, online purchase decision-making, and technology acceptance in digital retail environments [37,57]. Existing literature suggests that VTO technologies enhance shopping confidence by reducing uncertainty related to fit, style, and product appearance, thereby improving online purchase experiences. VTO is theorised as a higher order construct comprising two important sub-constructs based on the Technology Acceptance Model: Perceived Usefulness and Adoption Intention. Perceived Usefulness is used to explain how consumers believe that VTO will enhance their shopping performance by assisting them in visualisation of products more precisely, less uncertainty and make improved purchase decisions [19,37,45]. Adoption Intention illustrates the willingness and readiness of the consumer to make use of VTO technology when making online purchases [20,39]. Accuracy of Product Selection (APS) is the first mediator variable in this study. Accuracy of Product Selection refers to consumers' ability to select products that appropriately match their expectations regarding size, fit, and appearance during online shopping. The construct was developed based on prior studies related to fit prediction, product matching accuracy, and size recommendation systems in fashion e-commerce [31]. Existing studies demonstrate that uncertainty regarding fit and sizing significantly contributes to online product returns, while digital fitting technologies improve purchase confidence and reduce selection errors. APS is used to refer to the degree that consumers think they can choose the correct product on size, fit, and appearance in the case of VTO technology. Fashion retail online requires accurate selection because low accuracy is one of the primary reasons of products returns [11,15]. The dependent variable is Return Waste Reduction (RWR), namely reflects the decrease in product returns and environmental waste (including packaging waste) and transportation emissions, and reverse logistics burden [3,10,22]. Return Waste Reduction represents the extent to which VTO technologies help minimise unnecessary product returns, reverse logistics activities, packaging waste, and environmental burden associated with online shopping returns. The construct was adapted from sustainability and reverse logistics literature focusing on fashion e-commerce returns and environmental impacts [58]. Prior research highlights that return-related

logistics significantly increase carbon emissions, operational inefficiencies, and landfill waste within the online fashion industry. Lastly, there are Data Privacy Concerns that are considered as a moderating variable and are further classified into two dimensions, Negative Data Privacy Concerns (DPCN) and Positive Data Privacy Concerns (DPCP). DPCN is fear of misuse of data, surveillance, and no transparency and DPCP is trust in the practices of data handling of the platforms [20, 25–27,38]. Data Privacy Concerns reflects consumers' apprehensions regarding the collection, storage, sharing, and potential misuse of personal and biometric information while using VTO applications. The measurement items were adapted from prior privacy and security literature related to digital technologies, VTO systems, and privacy risk perceptions [57,58]. Existing studies indicate that VTO technologies frequently require facial images, body measurements, or biometric information, which may increase consumers' perceived privacy risks and negatively influence technology adoption behaviour.

2.4. Discussion of relationships among variables

The present study focuses on four major variables within the context of sustainable online fashion retailing: VTO, Accuracy of Product Selection (APS), Return Waste Reduction (RWR), and Data Privacy Concerns (DPC). VTO represents the use of digital and AR technologies that allow consumers to virtually visualise products before purchase. APS refers to consumers' ability to accurately evaluate product suitability in terms of size, fit, and appearance during online shopping. RWR represents the reduction of unnecessary product returns and associated environmental waste generated through reverse logistics and packaging activities. DPC reflects consumers' concerns regarding the collection, storage, and usage of personal and biometric information while using VTO technologies. Understanding the relationships among these variables is important in explaining how digital retail technologies contribute to both consumer decision-making and sustainable online shopping behaviour. It is believed that VTO technology will have a direct effect on the decrease of the waste in returns as the pre-purchase assessment of the consumers will be increased. Uncertainty that exists about size and fit is reduced as consumers are able to view the products virtually, which reduces the number of returns made after purchase. Previous research indicates that, immersive technologies enhance decision confidence and satisfaction that indirectly reduces the return behaviour [21,31]. Virtual Try-On (VTO) is also expected to have a positive effect on the accuracy of product selection (APS). When consumers perceive VTO as a useful technology and are willing to adopt it, they can evaluate products more effectively. The enhanced visual

realism and interactive features of VTO allow consumers to assess product fit, appearance, and suitability with greater precision, thereby improving the accuracy of their purchase decisions [15,37]. Greater accuracy in product selection is a critical factor in reducing product returns, as it minimizes dissatisfaction arising from discrepancies between consumer expectations and the actual product received. Since incorrect product selection and expectation–reality mismatches are among the most common reasons for online product returns, improving selection accuracy through VTO can contribute significantly to reducing return-related waste [13].

Therefore, APS is likely to become a major process in which VTO can affect return waste minimization. Nevertheless, these relationships could be weakened or strengthened based on the data privacy issues of consumers. Negative privacy issues are also high and this may make consumers shun the use of VTO completely hence undermining its usefulness. On the other hand, positive privacy attitudes can motivate more participation in VTO, which can reinforce its effects on long-term results.

2.5. Existing research gaps

There are still gaps in the research on VTO and online shopping behaviour despite the large body of research on the topic. First, the majority of the studies consider the consumer outcomes including purchase intention and satisfaction when only a little attention is paid to post-purchase and sustainability-related outcomes including the reduction of return waste. Second, accuracy of product choosing is described as the advantage of VTO, but it is hardly considered as a mediator variable. Third, the issue of data privacy is generally investigated as the direct predictor of adoption intention, but not as the moderator, which can determine the efficacy of technology on sustainability outcomes. Fourth, there are less studies that combine TAM and Privacy Calculus Theory to collectively explore the advantages of technology and privacy threats. These loopholes create a necessity to have a holistic model in terms of connection of VTO, the accuracy of selection, waste reduction of the returns, as well as privacy issues.

2.6. Relevant theories supporting the conceptual model

Technology Acceptance Model has given the basis on which the VTO experience, perceived usefulness, and adoption intention affect the quality of consumer decisions. TAM upholds the notion that the use of technology results in better performance of the tasks, which in this case is the accuracy of product selection. Privacy Calculus Theory is the counterpart to TAM since it describes the process of the consumer weighing the benefits against privacy risks. It justifies the moderating influence of the data privacy concerns in determining the effectiveness of VTO in reducing return waste.

2.7. Hypothesis development

2.7.1. VTO technology and reduced return waste

VTO is a technology that allows a consumer to digitally design an item in the form of AR and AI-based applications and remove the uncertainty when it comes to size, fit, and appearance. The initial computer-vision-driven VTO systems worked on enhancing the visual appearance and garment fit with a parser-free image processing method [59]. According to previous research, false expectations are a key source of returns of apparel, thereby contributing to the high logistical expenses and environmental waste [3,13]. Technologies of VTO help to visualize the product, direct the attention of the consumer, and optimize the online shopping experience. AR applications and realistic preview affect the purchase decision and engagement [60–62], and studies have suggested that their adoption and further optimization is increasing (Kumar, 2022). VTO reduces the expectation reality gap with its realistic previews, improving the level of purchase confidence and reducing the

probability of returns [33,37]. In view of the Technology Acceptance Model (TAM), consumers are more inclined to use technologies that they consider useful and this results in better decision-making results and fewer post-purchase dissatisfaction [44,45]. Since one of the most important sustainability issues in e-commerce fashion is return reduction, VTO will be positively involved in reducing the amount of returns waste. Therefore, it is hypothesised that:

H1: VTO technology is positively related to lower return waste significantly.

2.7.2. VTO technology and accuracy of product selection

Correctness of product choice will have to deal with how well the consumer is capable of picking the apparel based on the size, fit, and style expectations. VTO improves the accuracy of the selection by offering interactive visualization, multi-pose, and personalized representation H1: VTO technology is significantly positively associated with the decrease of the waste of returns [15,16]. AR-based VTO platforms take advantage of the automated material property extraction to create a more realistic product and increase user experience [63,64]. The development of image-based and deep learning-based VTO systems has increased the realism of garments and the accuracy of decisions even more [17,52]. TAM suggests that perceived usefulness will promote successful utilization of technology which ultimately enhances quality of decision making. It has been proven through empirically gathered evidence that consumers who use VTO have higher confidence and less perceived risk when they are otherwise online shopping clothing [21, 65]. With this, the adoption of VTO will show beneficial effects on the accuracy of the product selection. Therefore, it is hypothesised that:

H2: VTO technology is with the accuracy of product selection significantly positively related.

2.7.3. Accuracy of product selection and reduced return waste

In online fashion retailing, accurate product selection is a decisive factor of lowering the returns behaviour. Size and fit mismatches are also a common cause of returns and can be reduced when consumers make informed and accurate choices [11,12]. The enhanced accuracy of the selection reduces the rates of returns directly, and contributes to operational effectiveness and environmental sustainability [10,22]. This correlation is in line with TAM that implies that successful technology application results in significant behavioral changes. By consumers making the right product choice, both the return waste and textile disposal is reduced, as dissatisfaction goes down [53]. Therefore, it is hypothesised that:

H3: There is a significant positive correlation between the accuracy of product selection and less waste of returns.

2.7.4. Mediating role of accuracy of product selection

The effect of VTO on the reduction of return wastes is also indirect and works by enhancing accuracy in decision making. VTO is not a sufficient condition to decrease returns unless it increases the capacity of consumers to choose the appropriate goods [11,13]. The use of scenario-based tests has proven that VTO technology does enhance the perception of fitting the garment by a consumer, with less doubt and possible returns [66]. The empirical evidence indicates that VTO experiences that are immersive and personal lead to confidence in selection that, in turn, decreases return intentions [67,68]. TAM assists the given mechanism by stating that the perceived usefulness only converts to desirable effects when the technology is actually used in the course of decisions [44]. Therefore, it is hypothesised that:

H4: Product selection accuracy plays a significant mediating role in the relationship between VTO technology and low levels of returns wastage.

2.7.5. Moderating role of data privacy concerns

VTO systems are systems that demand the gathering of personal information including body measurements and visual information, which

could spell privacy problems among consumers. The privacy issues may be very high and restrict the use of VTO features, thus undermining their ability to enhance selection accuracy and minimize waste on returns [26,28]. Research on digital commerce has shown that privacy concerns have a harmful impact on the adoption and the intensity of technology use [20,35]. Meanwhile, the privacy calculus theory implies that the consumers assess the balance between perceived advantages and privacy threats. The less privacy concerns and with privacy concerns handled well, the consumers become more active on VTO and increase its effectiveness in reducing returns [27]. Therefore, it is hypothesised that:

H5: There are significant data privacy issues that moderate the relationship between VTO adoption and less return waste.

Based on the identified variables and theoretical relationships, the study proposes a conceptual framework in which VTO technology directly influences Return Waste Reduction and indirectly influences it through Accuracy of Product Selection. Furthermore, Data Privacy Concerns are proposed as a moderating factor affecting the effectiveness of VTO adoption in achieving sustainable shopping outcomes (see Fig. 1)

3. Methodology

3.1. Research design

This study adopted an explanatory sequential mixed-methods research design, integrating quantitative and qualitative approaches to obtain a comprehensive understanding of the relationship between VTO technology, product selection accuracy, return waste reduction, and data privacy concerns [69,70]. The quantitative aspect adheres to cross-sectional survey design that helps to obtain the structured information about a significant population of online shoppers at one time. This design would be appropriate in testing the relationship between the variables VTO experience, the correctness of product choice, reducing wastage, and privacy issues. The qualitative part is the semi-structured interviews, which will give more information about the experiences of the customers with VTO, their attitudes toward its privacy and awareness of sustainability. This design can facilitate the triangulation of results, increase the validity, as well as enable the study to include both statistical patterns, and real-life explanations. The study is explanatory because it aims to describe how and why VTO implementation has a role in reducing return wastes, with the accuracy of the product selection as a mediating factor, and the data privacy issues as the moderating factor. The quantitative phase was conducted first to statistically examine the hypothesized relationships among the study variables using a cross-sectional survey approach. The qualitative phase was subsequently conducted to explain and interpret the quantitative findings in

greater depth, particularly the behavioural reasons underlying VTO adoption, privacy concerns, and sustainability-related decision-making. For the qualitative phase, two participant groups were purposively selected to obtain contrasting yet complementary perspectives regarding online shopping behaviour and VTO usage. Group A consisted of frequent product returners, who were selected because they could provide deeper insights into the major causes of return behaviour, including issues related to sizing uncertainty, fit mismatch, and dissatisfaction with online product representation. Group B consisted of active VTO users, who were selected to understand how VTO technologies influence shopping confidence, product evaluation, return reduction behaviour, and privacy perceptions. The selection of these two groups enabled the study to compare pre-adoption and post-adoption behavioural experiences and provided richer contextual understanding of how VTO technologies influence sustainable online shopping practices.

The rationale for combining quantitative surveys with semi-structured interviews was to achieve methodological triangulation and enhance the validity of the findings. While the survey phase identified statistical relationships and behavioural patterns, the qualitative interviews provided contextual explanations regarding why consumers adopt VTO technologies despite privacy concerns and how VTO influences return-related behaviour in practice. This sequential design enabled both statistical validation and contextual interpretation of the research problem.

3.2. Sample profile

The target population of this study consisted of online fashion shoppers in India who had prior experience using VTO technologies on e-commerce platforms such as Myntra, Amazon Fashion, Ajio, and Nykaa Fashion. The target population of the study is young adults and working professionals because it is the most active group of online shopping and digital technology users [21]. In the case of the quantitative survey, 402 respondents were collected, which is deemed sufficient to use quantitative methods of data analysis and mediation-moderation testing [71]. The sample was comprised of students, salaried workers, and self-employed people with different frequencies of on-line shopping. Although the study primarily focused on respondents from Southern Karnataka, the online distribution approach enabled participation from consumers across nearby urban regions with active engagement in online fashion retailing. Therefore, the sample should be interpreted as digitally active online shoppers predominantly connected to the Southern Karnataka online retail ecosystem rather than being geographically restricted exclusively to one district or city. A purposive sampling technique was adopted to ensure that only respondents familiar with VTO technologies participated in the study. The

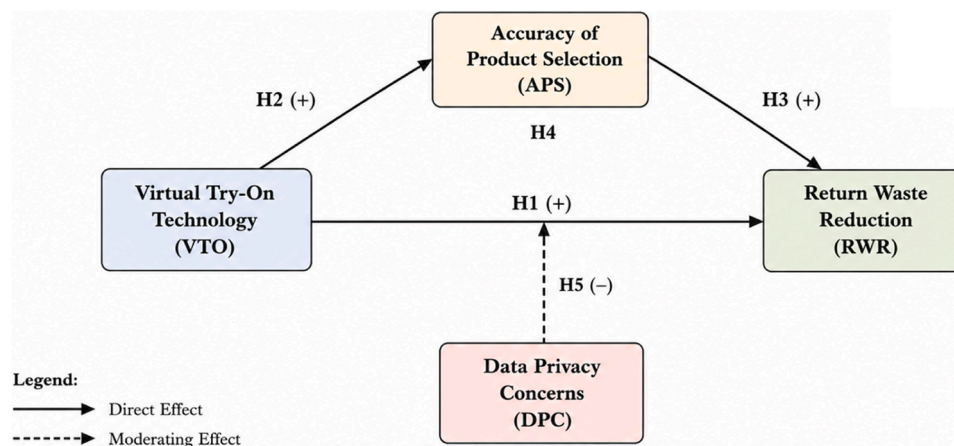


Fig. 1. Conceptual model.

Source: Compiled by the Author

qualitative sample was the smaller group of respondents chosen among the survey respondents willing to provide more detailed insights and this was done to guarantee the consistency between the quantitative and qualitative information and diversity in terms of gender, education level

and online shopping behavior.

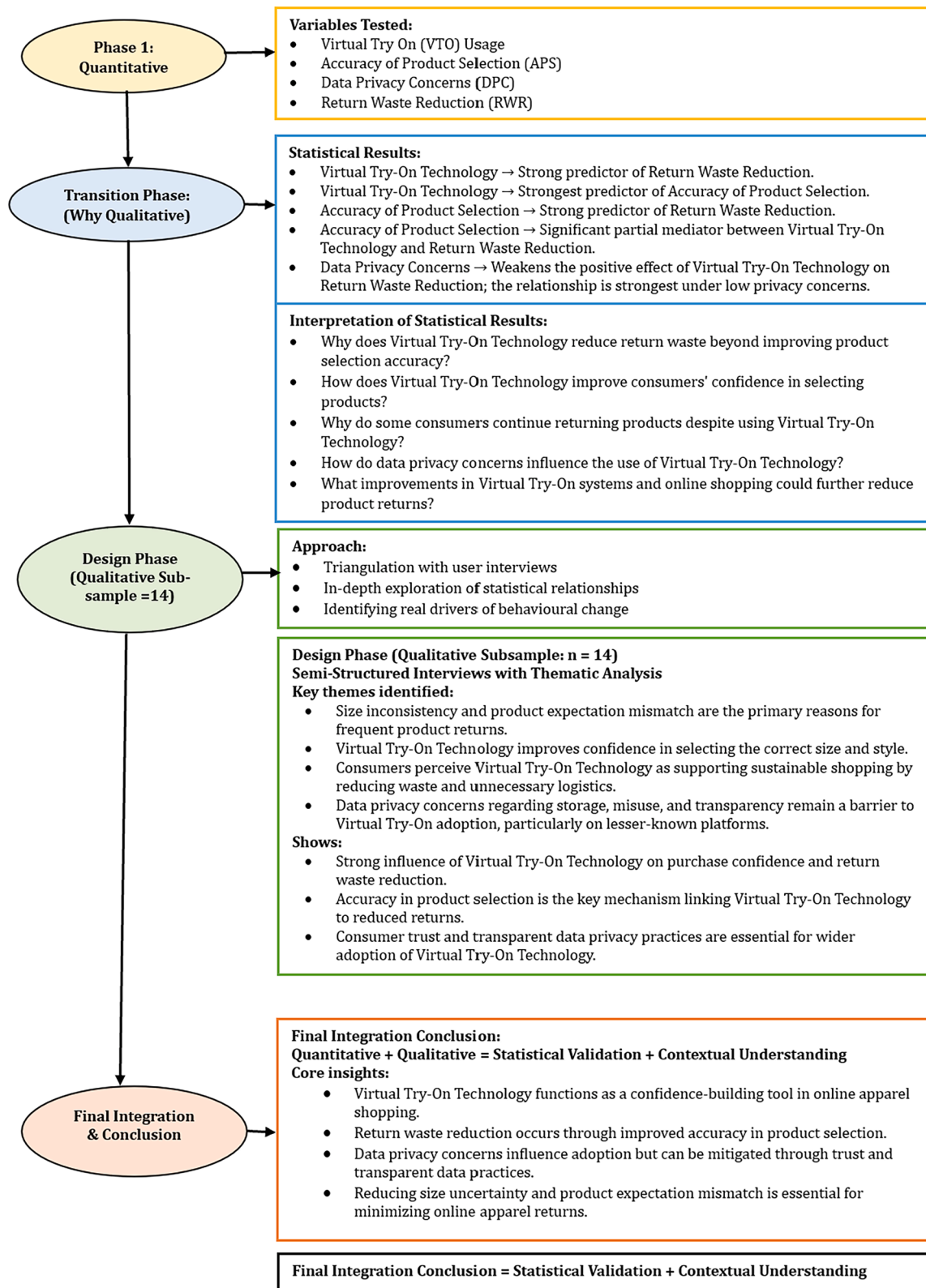


Fig. 2. Procedure of the study.
 Source: Compiled by the Author

3.3. Inclusion and exclusion criteria

In order to make the data relevant and of high quality, certain inclusion and exclusion criteria were used [72]. Participants were included in the study if they had previous experience with online shopping, had used VTO at least once, were aged 18–35 years or above, and gave informed consent to participate. Participants were excluded if they had never used any VTO or virtual fitting technology, provided incomplete or inconsistent survey responses, or did not agree to their data being used for academic research. These criteria were used to make sure the sample had informed users who can be able to make meaningful evaluation of VTO effectiveness and privacy issues.

3.4. Data collection and procedure

Primary data were collected in two sequential phases. In the first phase, a structured online questionnaire was distributed through multiple digital channels, including email, WhatsApp groups, Instagram, LinkedIn, and other social media platforms. Multiple distribution channels were intentionally used to improve respondent reach, diversify participant profiles, and increase access to active online shoppers familiar with digital retail technologies. Since the target respondents were digitally engaged consumers, online distribution was considered appropriate for accessing individuals with prior VTO experience.

Before participation, respondents were informed about the study objectives, confidentiality measures, voluntary participation, and academic purpose of the research. Responses were screened to remove incomplete, duplicate, and inconsistent submissions to maintain data quality and reliability.

The second phase involved semi-structured interviews with selected respondents from the survey phase. The purpose of this qualitative stage was not merely exploratory but explanatory in nature, aiming to clarify the behavioural reasoning behind the quantitative findings. Specifically, the interviews investigated why consumers continued using VTO technologies despite privacy concerns, why sustainability orientation alone did not strongly predict return reduction behaviour, and how trust, convenience, and perceived usefulness influenced technology adoption. Interviews were conducted either online or face-to-face depending on participant convenience, recorded with consent, and transcribed for thematic analysis.

Fig. 2 presents the sequential methodological process adopted in the study, beginning with quantitative hypothesis testing followed by qualitative interpretation and final integration of findings.

3.5. Research instrument

The present study employed both quantitative and qualitative research instruments to comprehensively examine consumer perceptions, adoption behaviour, sustainability outcomes, and privacy concerns associated with VTO technology in online fashion retailing. The instruments were developed based on previously validated scales from existing literature and were adapted to suit the context of online fashion shopping and VTO usage.

3.5.1. Quantitative research instrument

The quantitative instrument consisted of a structured questionnaire divided into multiple sections corresponding to the major constructs of the study. All measurement items were assessed using a five-point Likert scale ranging from 1 = “Strongly Disagree” to 5 = “Strongly Agree.” The questionnaire was designed to capture respondents’ perceptions, attitudes, and behavioural intentions regarding VTO technology, return reduction behaviour, product selection accuracy, and data privacy concerns.

The construct of VTO measured respondents’ overall perceptions regarding the usefulness, usability, confidence, and effectiveness of VTO systems during online shopping. The items were adapted primarily from

studies related to AR shopping technologies and consumer responses toward VTO applications [57]. Sample items included perceptions regarding shopping accuracy, confidence in purchase decisions, and recommendation intention toward VTO usage.

Perceived Usefulness (PU) assessed the extent to which respondents believed that VTO systems improve shopping convenience, enhance purchase decisions, and reduce return-related uncertainty. These items were adapted from the Technology Acceptance Model-related literature examining technology adoption and perceived usefulness dimensions [37]. The construct included items associated with shopping convenience, size accuracy, and overall enhancement of online shopping experience.

Adoption Intention (AI) measured consumers’ willingness to use VTO technologies during future online shopping activities. The scale evaluated respondents’ likelihood of continued usage, recommendation intention, and reliance on VTO systems before final purchase decisions. The items were also adapted from prior technology adoption studies [37].

Return Waste Reduction (RWR) examined consumers’ perceptions regarding the ability of VTO technologies to minimise unnecessary product returns, shipping waste, environmental impact, and unsustainable shopping practices. The items were adapted from sustainability and waste reduction literature associated with green technology adoption and sustainable consumption behaviour [31].

Accuracy of Product Selection (APS) measured the extent to which VTO systems improve consumers’ confidence in selecting appropriate products based on size, fit, appearance, and style. The items were adapted from studies examining AI-enabled customer decision support systems and online purchase confidence [31,58]. This construct captured respondents’ confidence in product matching and reduction of incorrect purchase decisions.

Data Privacy Concerns (DPC) assessed respondents’ concerns regarding biometric data collection, sharing of personal information, and privacy risks associated with VTO technologies. The measurement items were adapted from privacy and online disclosure literature focusing on technology-mediated consumer interaction [57]. The items examined respondents’ discomfort with sharing sensitive information and concerns regarding third-party access to personal data.

Additionally, Positive Privacy Perception (PP) measured respondents’ trust in VTO platforms regarding the responsible handling of personal information, the transparency of privacy policies, and their willingness to share data for an enhanced shopping experience. The items were adapted from prior studies related to green technology and technology-enabled consumer trust [31].

Before final data collection, the questionnaire underwent expert review and pilot testing to ensure clarity, relevance, and contextual appropriateness of the measurement items. Minor wording modifications were incorporated based on expert suggestions and participant feedback.

3.5.2. Qualitative research instrument

The qualitative instrument consisted of semi-structured interview questions designed to obtain deeper insights into consumer experiences, behavioural attitudes, return behaviour, and privacy perceptions associated with VTO technologies. Two separate interview groups were considered: (1) online shoppers who frequently return products, and (2) online shoppers who actively use VTO technologies.

The interview questions for frequent returners focused on identifying the major reasons behind product returns, trust toward online shopping, environmental awareness regarding return-related waste, and perceptions regarding technologies capable of reducing return behaviour.

The interview questions for VTO users explored first-hand experiences with VTO systems, confidence in selecting products, reduction of return behaviour, satisfaction with online shopping experiences, and concerns regarding sharing body measurements or personal images while using VTO applications.

The qualitative instrument enabled the study to capture behavioural explanations and contextual insights that complemented the quantitative findings, thereby strengthening the overall validity and depth of the mixed-methods research design.

3.6. Reliability and validity checks

To determine reliability and validity, the research findings were required to be robust [71,72]. Cronbach alpha was used to evaluate internal consistency reliability with all the constructs having values exceeding the cut-off point of 0.70. Expert review and changing the measurement items of other known literature were the methods used to guarantee content validity. Factor analysis was applied to determine construct validity by ensuring that the items were loading well on the constructs. The validity of the study was also increased through triangulation of quantitative and qualitative results because it triangulated the outcomes with other results [11,73].

3.7. Analytical techniques

Quantitative data were analysed using IBM SPSS Statistics and SmartPLS 4 software. SPSS was used for preliminary data screening, descriptive statistics, demographic profiling, and correlation analysis. SmartPLS 4 was employed to conduct Partial Least Squares Structural Equation Modelling (PLS-SEM), as the study involved mediation and moderation analysis with multiple latent constructs [74].

The measurement model was evaluated using indicator reliability, composite reliability, Cronbach's alpha, Average Variance Extracted (AVE), and discriminant validity measures. The structural model was subsequently assessed through path coefficients, coefficient of determination (R^2), predictive relevance (Q^2), effect size (f^2), and bootstrapping procedures to test the proposed hypotheses. Mediation analysis was conducted to examine the mediating role of Accuracy of Product Selection (APS), while moderation analysis tested the moderating effect of Data Privacy Concerns (DPC) on the relationship between VTO and Return Waste Reduction (RWR).

Following the quantitative phase, qualitative interview data were analysed using thematic analysis. Interview transcripts were coded and categorised into themes associated with perceived usefulness, decision confidence, privacy concerns, sustainability orientation, and trust toward VTO technologies. The integration of quantitative and qualitative findings enabled deeper interpretation of the behavioural mechanisms underlying the statistical relationships identified in the study.

4. Analysis and results

4.1. Sample characteristics

Among 402 participants, most of them were between 18–35 (74.4%), which means that the sample included young and technologically active online shoppers who are more inclined to use the VTO technology. The proportion of female respondents was 61.4 percent ($N = 247$) which indicates the prevalence of women in online shopping in clothing. The majority of the respondents were graduates (38.3) and postgraduates (37.8) in the questionnaire, indicating a high level of the respondents who were well educated to comprehend the issue of technology use, sustainability, and data privacy. The greatest population (36.1 and 27.9) are working professionals and students respectively, which means they interact with online shopping platforms regularly. Most of the respondents were between low to middle-income groups, which emphasized the need to make the right choice of product to prevent unwarranted returns. The majority of participants lived in towns and cities, and online shopping was very frequent, which allows considering the sample appropriate to the study of VTO adoption in online fashion shopping (see Table 2)

Table 2
Sample and background profile.

Demographics	Category	Number of Respondents	Percentage
Gender	Male	155	38.6%
	Female	247	61.4%
Age (Years)	18–25	145	36.1%
	26–33	154	38.3%
	33–40	76	18.9%
	41 & above	27	6.7%
Education Level	High School	17	4.2%
	PUC	46	11.4%
	Undergraduate	154	38.3%
	Postgraduate	152	37.8%
	Doctorate	22	5.5%
	Diploma	11	2.7%
Occupation	Student	112	27.9%
	Working Professional	145	36.1%
	Self-Employed	107	26.6%
	Homemaker	29	7.2%
	Unemployed	9	2.2%
Monthly Income (INR)	Below 10,000	67	16.7%
	10,000–25,000	178	44.3%
	25,001–50,000	113	28.1%
	50,000 & above	44	10.9%
Online Shopping Frequency	Rarely (<1/month)	65	16.2%
	Occasionally (1–2/month)	157	39.1%
	Frequently (3–5/month)	161	40.0%
	Very Frequently (>5/month)	19	4.7%
Online Shopping Expenditure (INR)	<1,000	48	11.9%
	1,000–5,000	237	59.0%
	5,001–10,000	95	23.6%
	10,001 & above	22	5.5%
Platforms Used for VTO	Myntra	158	39.3%
	Amazon	209	52.0%
	Ajio	163	40.5%
	Nykaa Fashion	90	22.4%
	Other	12	3.0%
Concerns While Using VTO	Virtual fit inaccurate	67	16.7%
	Product looks different	138	34.3%
	Privacy concerns	250	62.2%
	VTO not available for all products	84	20.9%
Frequency of Product Returns	Never returned	21	5.2%
	Rarely	163	40.5%
	Occasionally	157	39.1%
	Frequently	61	15.2%
Willingness to Share Personal Data	Very willing	67	16.7%
	Somewhat willing	133	33.1%
	Neutral	168	41.8%
	Unwilling	23	5.7%
	Strongly unwilling	11	2.7%

Source: Authors' Analysis

4.2. Data description and validation

To ensure statistical validity and avoid reliance on simple arithmetic averages, the constructs were further examined using Principal Component Analysis (PCA) with Varimax rotation. PCA was employed to identify the underlying factor structure of the measurement items and to account for differing item contributions across constructs. The suitability of the data for factor analysis was first assessed using the Kaiser–Meyer–Olkin (KMO) Measure of Sampling Adequacy and Bartlett's Test of Sphericity.

The KMO value was 0.935, which exceeds the recommended threshold of 0.70, indicating excellent sampling adequacy for factor analysis. Bartlett's Test of Sphericity was statistically significant ($\chi^2 = 4415.498$, $df = 210$, $p < 0.001$), confirming that the correlation matrix was appropriate for factor extraction.

The PCA results extracted four principal components with

eigenvalues greater than 1, collectively explaining 64.294% of the total variance. The first component explained 44.540% of the variance, followed by the second (9.254%), third (5.795%), and fourth (4.705%) components. These results indicate that the retained components adequately captured the underlying dimensionality of the constructs used in the study.

The rotated component matrix further demonstrated that the measurement items loaded appropriately onto their respective constructs, confirming construct validity. Items associated with VTO usefulness, adoption intention, product selection accuracy, sustainability orientation, and privacy concerns exhibited strong factor loadings above the acceptable threshold of 0.50. Specifically, items related to confidence in product selection, reduction of wrong fit selection, and intention to rely on VTO before purchase showed particularly strong loadings, indicating that these variables are central dimensions underlying VTO adoption behaviour (see Table 3).

4.2.1. Virtual Try-On (VTO)

Typically, VTO technology is viewed by respondents favourably. The greatest mean was found with VTO assisting in visualizing products prior to purchase ($M = 3.89$, $SD = 0.91$), and then with increasing confidence in online shopping ($M = 3.74$, $SD = 0.96$). The ease of using VTO was a little less ($M = 3.42$, $SD = 1.04$), which says that there are some slight usability issues. The t-tests of independent samples proved the significance of all items: VTO1 ($t = 60.402$, $p < 0.001$), VTO2 ($t = 94.226$, $p < 0.001$), VTO3 ($t = 79.476$, $p < 0.001$), VTO4 ($t = 69.861$, $p < 0.001$), and VTO5 ($t = 62.678$, $p < 0.001$). On the whole, mean score ($M = 3.6264$, $SD = 0.77221$) and t-test results show that the perceived benefits of visualization and confidence associated with VTO are highly accepted.

4.2.2. Accuracy of Product Selection (APS)

VTO was discovered to be of great help in improving product selection accuracy. There was a strong agreement in the fact that it improves selection of size and fit ($M = 4.12$, $SD = 0.87$) and confidence in selecting the styles ($M = 3.96$, $SD = 0.90$), and confidence in colour and texture scored slightly lower ($M = 3.68$, $SD = 0.98$). t-tests showed significance in all items: APS1 ($t = 71.210$, $p < 0.001$), APS2 ($t = 85.412$). The averaged APS ($M = 3.6527$, $SD = 0.77092$) evidences the fact that VTO enhances the accuracy of product selection.

4.2.3. Return Waste Reduction (RWR)

The respondents relate the use of VTO to fewer product returns and environmental-friendliness. The most significant agreement was with the correct choice of reducing returns ($M = 3.94$, $SD = 0.93$), and awareness of the environmental impact was moderate ($M = 3.51$,

$SD = 1.02$). t-tests showed that all the items were significantly different (RWR1 ($t = 78.651$, $p < 0.001$), RWR2 ($t = 96.892$, $p < 0.001$), RWR3 ($t = 76.485$, $p < 0.001$), RWR4 ($t = 76.485$, $p < 0.001$), RWR5 ($t = 76.485$, $p < 0.001$)). Mean ($M = 3.6264$, $SD = 0.77221$) shows that the respondents think that VTO is effective to minimize return waste.

4.2.4. Data Privacy Concerns (DPC) positive aspect

Favorable views of information privacy enhance VTO participation. The respondents agreed with the statements that clear privacy policies make them more willing to use VTO ($M = 4.05$, $SD = 0.88$) and that secure platforms make them more confident ($M = 3.91$, $SD = 0.92$). t-tests found that they were significant: DPCP1 ($t = 77.404$, $p < 0.001$), DPCP2 ($t = 88.593$, $p < 0.001$), DPCP3 ($t = 71$ Mean ($M = 3.6592$, $SD = 0.68732$) indicates high trust and positive perceptions of privacy.

4.2.5. Data Privacy Concerns (DPCN) negative aspect

Adoption of VTO may be restricted by negative privacy issues. There was a discomfort in sharing personal data ($M = 3.75$, $SD = 1.173$), fears of third-party access ($M = 3.66$, $SD = 0.845$) with significance: DPCN1 ($t = 64.013$, $p < 0.001$), DPCN2 ($t = 86.934$, $p < 0.001$), DPCN3 ($t = 65.286$, $p < 0.001$), DPC Total mean ($M = 3.5572$, $SD = 0.76400$) proves that there is a fear of privacy but all has not been completely negative regarding VTO benefits.

4.3. Measurement model evaluation

To test the reliability and validity of the constructs employed in the research, the measurement model was assessed in terms of the internal consistency reliability, convergent validity, discriminant validity, multicollinearity diagnostics, as well as the overall model fit measures, according to the general criteria of PLS-SEM. During the measurement model assessment and Principal Component Analysis (PCA), several items from the VTO, Accuracy of Product Selection (APS), and Return Waste Reduction (RWR) constructs were excluded due to low factor loadings and weaker contribution to construct reliability and validity. Only items with acceptable loading values above the recommended threshold were retained for the final analysis to ensure convergent validity and construct consistency. For the VTO construct, seven items (VTO2, VTO4, VTO5, PU2, PU4, AI4, and AI5) were retained, with factor loadings ranging from 0.631 to 0.772. Similarly, within the Accuracy of Product Selection construct, three items (APS3, APS4, and APS5) demonstrated satisfactory loadings ranging from 0.713 to 0.843 and were therefore retained. For Return Waste Reduction, three items (RWR3, RWR4, and RWR5) were retained, with factor loadings ranging between 0.653 and 0.746. Items exhibiting comparatively lower loadings, cross-loadings, or weaker construct contribution were removed to improve Composite Reliability (CR), Average Variance Extracted (AVE), and overall measurement model quality, consistent with recommended guidelines for PLS-SEM analysis.

Cronbach alpha and composite reliability (ρ_c) were used in measuring internal consistency reliability (see Table 4). The findings show that there is a satisfactory reliability in all constructs since the values of all constructs were above the recommended level of 0.70. Product Selection Accuracy reported Cronbach alpha of 0.824 and composite reliability of 0.827, Return Waste Reduction had a Cronbach alpha of 0.751 and 0.755 and VTO had a Cronbach alpha of 0.882 and composite reliability of 0.884. These findings indicate that the measurement items are always consistent in terms of their constructs. Average Variance Extracted (AVE) was used to test convergent validity. The constructs were all found to have AVE that exceeded the minimum value of 0.50 thus showing acceptable convergence. The accuracy of Product Selection registered an AVE of 0.616, Return Waste Reduction registered an AVE of 0.506, and VTO registered an AVE of 0.517. This confirms that the explanation of each construct accounts to over 50 percent of the variance of its indicators, which prove the suitability of the model of measurement.

The Heterotrait-Monotrait (HTMT) ratio was used to determine the

Table 3

Descriptive statistics of each variable.

Construct	No. of Items	Mean	SD	Dominant Loading Range	PCA Component
VTO Perception	5	3.626	0.772	0.575–0.760	Component 1
Usefulness of VTO	5	3.624	0.761	0.560–0.729	Component 1
Adoption Intention	5	3.645	0.750	0.580–0.728	Component 1
Return Waste Reduction	5	3.626	0.772	0.597–0.695	Component 2
Accuracy of Product Selection	5	3.653	0.771	0.676–0.760	Component 3
Negative Privacy Concerns	4	3.557	0.764	0.565–0.945	Component 4
Positive Privacy Perception	4	3.659	0.687	0.520–0.728	Component 4

Source: Authors' Analysis

Table 4

Mean, S.D., outer loadings, reliability and validity results.

Construct	Sources	Item Code	Description	Mean	S.D.	λ	CR	AVE
Virtual Try-On (VTO)	[57] [75]	VTO2	I understand how to use VTO without much effort	3.82	0.812	0.631	0.882	0.517
		VTO4	I would recommend VTO to others for online shopping	3.54	1.016	0.712		
		VTO5	VTO accurately represents how a product would look on me	3.41	1.091	0.763		
		PU2	The VTO feature would make online shopping for clothes more convenient for me	3.75	0.864	0.705		
		PU4	The VTO feature would reduce the chances of returning products due to wrong fit	3.45	1.082	0.748		
		AI4	I would recommend using the VTO feature to others who shop for apparel online	3.53	1.019	0.695		
Accuracy in Product Selection	Hayat, F. A. (2025)	AI5	I am likely to rely on the VTO feature before making my final purchase decision	3.40	1.099	0.772	0.824	0.616
		APS3	Using VTO reduces the chance of choosing the wrong size or fit	3.57	1.085	0.713		
		APS4	VTO improves my ability to assess how a style will look on me	3.58	1.021	0.793		
Return Waste Reduction (RWR)	Li, C. et al. (2023)	APS5	I feel confident that my product choice is accurate after using VTO	3.48	1.099	0.843	0.751	0.506
		RWR3	I believe VTO is an effective tool for making online shopping more environmentally friendly	3.69	0.966	0.653		
		RWR4	I prefer to shop from online retailers that implement technologies like VTO to reduce environmental impact	3.53	0.974	0.731		
		RWR5	Using VTO aligns with my personal values regarding environmental responsibility in consumption	3.51	1.046	0.746		

Source: Authors' Analysis

discriminant validity (see Table 5). The values of all the HTMTs were within acceptable ranges, which proves that the constructs are empirically different. Even though somewhat high HTMTs were found between VTO and Accuracy of Product Selection (HTMT = 0.894) and between VTO and Return Waste Reduction (HTMT = 0.882), they were not above the recommended one. This implies a close relationship between the constructs which are conceptually distinct and hence, satisfactory discriminant validity.

The Multicollinearity was evaluated through the Variance Inflation Factor (VIF). The VIF of the outer model values had a range of 0.812 to 0.831 indicating they were very low considering the recommended value of 5. It means that the indicators do not have a problem of multicollinearity. Also, the bootstrap confidence intervals (2.5% 97.5) were between 0.761 and 0.868 which also confirmed the stability and strength of the parameter estimates. Model fit was assessed on the basis of world fit indices. The value of Standardized Root Mean square residual (SRMR) of the saturated and estimated models was 0.09, which is a good fit. Other fit statistics, such as d_{ULS} (1.238), d_G (0.465), chi-square (992.931), and Normed Fit Index (NFI = 0.752), indicate that the presented measurement model appropriately describes the data it was optimized on and can be further examined structurally. The measure of the indicator reliability was determined by outer loadings.

The best loadings of the VTO construct (0.631 to 0.763) with VTO5 contributing the most (0.763) were observed, which showed that VTO5 made significant contributions to the construct. Adoption Intention The conceptual relevance of two items (AI4 (0.695) and AI5 (0.772)) was used to measure Adoption Intention. There were high loadings of PU2 (0.705) and PU4 (0.748), which confirmed the perception of the respondents about the usefulness of VTO. Return Waste Reduction construct recorded a load between 0.653 and 0.746, whereas Accuracy of Product Selection had a high loading of between 0.713 and 0.843 with

APS5 load being the highest (0.843).

All in all, measurement model shows a high level of reliability, good convergent and discriminant validity, no multicollinearity, and reasonable model fit. These findings support the argument that constructs and indicators utilized in the analysis are psychometrically viable and were appropriate to test hypotheses and structural relationships proposed.

4.4. Structural model results

Fig. 3.

The results of the structural model show that VTO has a direct and indirect impact on Return Waste Reduction (RWR). Table 6 demonstrates that the hypothesized links are supported as determined through direct effect analysis, with positive and significant relationships between VTO and RWR ($\beta = 0.394$, $t = 6.242$, $p < 0.001$, H1), VTO and Accuracy of Product Selection (APS) ($\beta = 0.827$, $t = 41.407$, $p < 0.001$, H2), and APS and RWR ($\beta = 0.506$, $t = 7.894$). The mediation (Table 7) results confirm that APS is a contributing factor to the VTO-RWR relationship (indirect effect = 0.418, total effect = 0.812, $p < 0.001$, H4), and objectively selecting products is one of the factors that reduce returns. Data Privacy Concerns Negative (DPCN) moderates the VTO RWR relationship (value of -0.038 and significance of 1.4) whereas Data Privacy Concerns Positive (DPCP) does not (value of -0.028 and significance of 0.2) as seen in the moderation analysis (Table 8). Bootstrapped histograms of direct and total effects have bell-shaped shapes, meaning they provide stable, high-quality and consistent estimates, and therefore VTO has a significant effect on reducing the wastes related to returns both directly and through increased accuracy in picking products, which supports the proposed model.

4.5. Qualitative results

The qualitative results are based on in-depth interviews with online apparel shoppers, coded from A–N and divided into two groups: Group A (frequent returners) and Group B (VTO users). The analysis helped identify patterns that support the quantitative findings and give better understanding of consumer behaviour. Participants in Group A said they often returned products because of size and fit issues, differences between the product shown online and the actual product, and frustration during online shopping, which made them order multiple sizes or options. On the other hand, participants in Group B had positive

Table 5

Heterotrait–Monotrait (HTMT).

Variable	Accuracy in Product Selection	Return Waste Reduction	Virtual Try On
Accuracy in Product Selection			
Return Waste Reduction	0.849		
Virtual Try On	0.844	0.842	

Source: Authors' Analysis

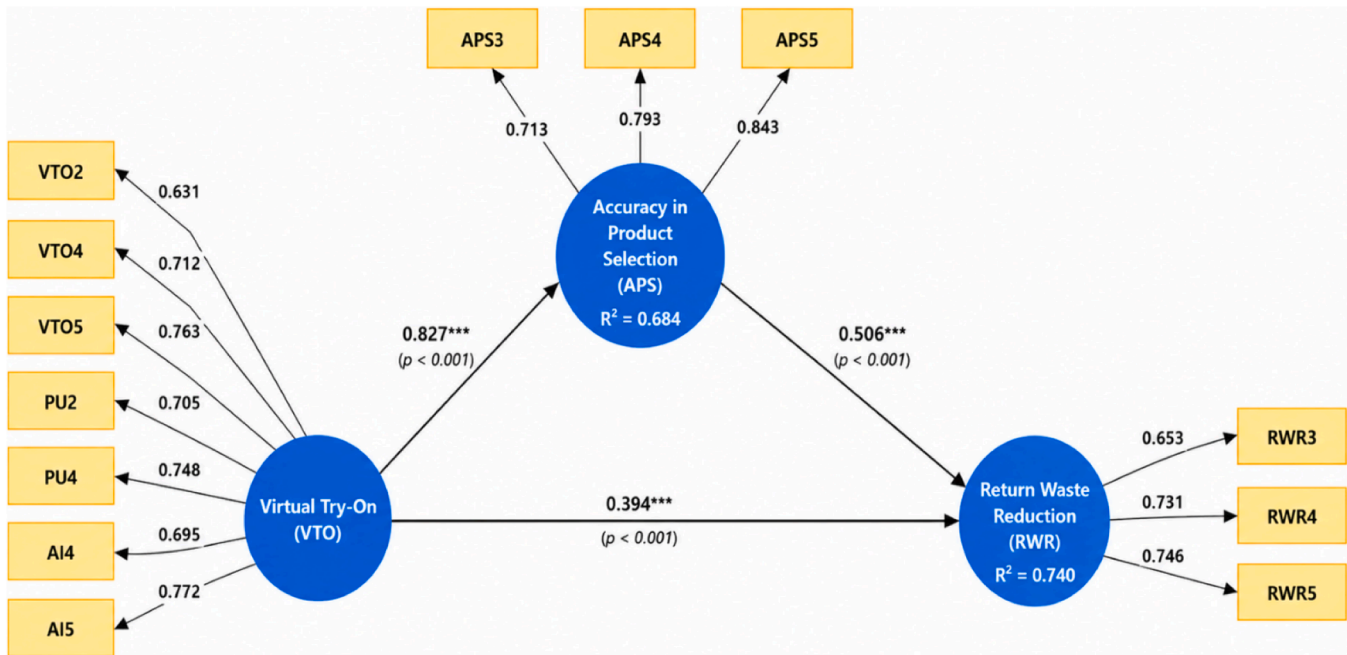


Fig. 3. Structural model.
Source: Authors' Analysis – restructured from SmartPLS output.

Table 6
Direct effect.

H	Direct Effect	β	t-Statistics	p-values	Decision
H1	VTO $\rightarrow$ RWR	0.394	6.242	0.000	Supported
H2	VTO $\rightarrow$ APS	0.827	41.407	0.000	Supported
H3	APS $\rightarrow$ RWR	0.506	7.894	0.000	Supported

Source: Authors' Analysis

experiences with VTO. They said it was easy to use and helped them feel more confident in choosing the right size and style, which reduced unnecessary returns. However, some participants also shared concerns about sharing personal or biometric data and said they prefer platforms that clearly explain how their data is used.

The combined quantitative and qualitative discussion (see Table 9) reveals that VTO technology has a strong impact on the online shopping behaviour of clothing. Returning customers were uncertain about size, fit, and appearance, and thus dissatisfied with that, which is in line with the quantitative relationship between uncertainty in fit and return intention. Contrary to this, the active users of VTO reported a greater level of confidence, satisfaction, and reduced returns because VTO allows visualizing the correct fit and minimizing decision anxiety. This helps in sustainable consumption, since there is reduced unnecessary returns. All in all, VTO fills the gap between the demands and results of the product, yet to win trust and make people more willing to adopt it, the issue of data privacy needs to be addressed with clear policies.

5. Discussion

The present study examined the role of VTO technology in improving product selection accuracy and reducing return-related waste within

online fashion retailing while considering the moderating influence of data privacy concerns. The findings provide strong empirical support for the integrated conceptual framework developed using the Technology Acceptance Model and Privacy Calculus Theory. Specifically, the results demonstrate that consumers are more likely to adopt and positively evaluate VTO technologies when they perceive them as useful in reducing shopping uncertainty, improving confidence, and facilitating more accurate purchase decisions. These findings are consistent with the assumptions of the Technology Acceptance Model (TAM), which suggests that perceived usefulness significantly influences behavioural intention and technology adoption [40].

The structural model results revealed that VTO Technology has a

Table 8
Moderation effect.

Hypothesis	Relation	Data Privacy Concerns					
		High		Moderate		Low	
		Beta	P	Beta	P	Beta	P
H5	Virtual Try On * Data Privacy Concerns Positive $\rightarrow$ Return Waste Reduction	.247	0.004	.477	0.000	.786	0.000
	Virtual Try On * Data Privacy Concerns Negative $\rightarrow$ Return Waste Reduction	.423	0.000	.720	0.000	.671	0.000

Source: Authors' Analysis

Table 7
Mediation effect.

H	Path	DE	TE	p-value	Path	IE	p-value	Decision
H4	VTO $\rightarrow$ RWR	0.394	0.812	0.000	VTO $\rightarrow$ APS $\rightarrow$ RWR	0.418	0.000	Supported

Source: Authors' Analysis

Table 9
Integrated analysis of quantitative and qualitative results.

Theme	Quantitative Findings	Qualitative Findings	Integrated Interpretation
Uncertainty in Size and Fit	High perceived fit risk and size uncertainty were found to significantly increase return intention among online shoppers.	Group A participants reported confusion due to inconsistent size charts and brand variations, leading to frequent returns.	Both findings indicate that lack of fit confidence is a major reason for high return behaviour in online apparel shopping.
Product Expectation Mismatch	Visual mismatch between product images and actual items was positively associated with dissatisfaction and returns.	Participants from Group A highlighted differences in colour, fabric texture, and overall appearance compared to online images.	The combined results confirm that unrealistic online representations increase dissatisfaction and return rates.
Confidence in Purchase Decision	VTO usage showed a significant positive impact on purchase confidence and decision accuracy.	Group B participants stated that VTO helped them visualise fit and style clearly, reducing hesitation and second guessing.	The qualitative insights explain how VTO improves confidence, supporting the quantitative relationship observed.
Reduction in Product Returns	VTO adoption was negatively related to return intention and return frequency.	Group B participants shared multiple instances where VTO helped them avoid ordering incorrect sizes or unsuitable styles.	Both results strongly support that VTO is effective in reducing apparel returns.
Sustainable and Mindful Shopping Behaviour	VTO usage was positively associated with reduced over-ordering and sustainable shopping behaviour.	Participants acknowledged that VTO helped them make informed choices, avoid waste, and reduce unnecessary packaging and logistics.	The integrated findings highlight VTO as a tool that supports environmentally responsible consumption.
Data Privacy and Trust Concerns	Privacy concerns were identified as a moderating factor affecting VTO adoption.	Group B participants expressed concerns regarding data storage, misuse, and transparency, especially on lesser-known platforms.	While VTO benefits are clear, both findings stress that data privacy and ethical data practices are crucial for long-term adoption.

Source: Compiled by the Author

significant positive influence on Accuracy of Product Selection (APS) ($\beta = 0.827$, $t = 41.407$, $p < 0.001$), thereby supporting H2. This finding strongly reinforces the TAM-based argument proposed in the conceptual framework that consumers adopt VTO systems because they perceive them as useful in improving online purchase decision-making. The significantly high path coefficient indicates that VTO technologies substantially improve consumers' ability to evaluate product fit, size, style, and appearance before purchase. These findings align with earlier studies showing that AR and VTO applications improve online shopping confidence and reduce purchase ambiguity by enabling visually supported product evaluation [13,37]. Furthermore, the qualitative findings provide contextual support for this relationship, as participants from Group B reported that VTO systems reduced hesitation, improved fit visualisation, and enhanced confidence in selecting appropriate apparel products.

The findings additionally confirmed that Accuracy of Product

Selection significantly influences Return Waste Reduction ($\beta = 0.506$, $t = 7.894$, $p < 0.001$), thereby supporting H3. This result demonstrates that accurate product evaluation and confident purchase decisions substantially reduce unnecessary product returns and associated environmental waste. The result directly supports the sustainability-oriented assumptions proposed in the conceptual model, where APS functions as a behavioural mechanism linking technology adoption with sustainable retail outcomes. Previous studies similarly identified fit uncertainty, inaccurate product expectations, and mismatch between online product representation and actual products as major causes of return behaviour in fashion e-commerce [11,32]. The qualitative findings further strengthen this relationship, as frequent returners in Group A described confusion regarding sizing inconsistency, product mismatch, and dissatisfaction with actual product appearance, which frequently resulted in over-ordering and repeated returns. Thus, the present study confirms that improving decision accuracy through VTO technology can substantially minimise reverse logistics activities, packaging waste, transportation emissions, and unsustainable return practices.

The direct effect analysis further showed that VTO Technology significantly influences Return Waste Reduction ($\beta = 0.394$, $t = 6.242$, $p < 0.001$), thereby supporting H1. This finding demonstrates that VTO technologies contribute directly to environmentally sustainable shopping behaviour beyond their immediate technological utility. Consumers using VTO systems are less likely to engage in impulsive purchases, multiple-size ordering, or uncertainty-driven return behaviour. This finding is consistent with prior sustainability-oriented research suggesting that digital fitting technologies and AI-assisted shopping tools reduce return-related inefficiencies [65,76]. The qualitative evidence also supports this relationship, as VTO users reported fewer return experiences, greater satisfaction with purchases, and more mindful shopping behaviour after using VTO systems.

Importantly, the mediation analysis confirmed that the accuracy of product selection partially mediates the relationship between VTO Technology and Return Waste Reduction, supporting H4. The indirect effect was found to be significant ($\beta = 0.418$, $p < 0.001$), while the total effect of VTO on Return Waste Reduction was $\beta = 0.812$. This finding provides deeper theoretical explanation regarding how VTO contributes to sustainability outcomes. Specifically, VTO technologies reduce return-related waste not merely through technological presence alone, but because they improve consumers' decision-making accuracy during online shopping. In other words, consumers who are able to better visualise and evaluate products before purchase are less likely to engage in unnecessary return behaviour. This mediation relationship strengthens the applicability of the Technology Acceptance Model by demonstrating that perceived usefulness translates into sustainability outcomes through enhanced consumer judgement and behavioural confidence. Similar findings were reported by [33,65], who found that technology-enabled purchase confidence significantly reduces post-purchase dissatisfaction and return behaviour.

The moderation analysis further strengthens the relevance of Privacy Calculus Theory within the study context. The findings revealed that negative Data Privacy Concerns significantly moderate the relationship between VTO and Return Waste Reduction, while positive privacy perceptions strengthen technology acceptance and sustainable usage behaviour. This indicates that consumers evaluate VTO technologies by balancing perceived technological benefits against perceived privacy risks. Although consumers recognised the usefulness and convenience of VTO systems, concerns regarding biometric data collection, facial recognition, image storage, and third-party access to personal information negatively influenced trust and technology engagement. These findings are consistent with recent privacy-oriented VTO studies showing that lack of transparency and concerns regarding misuse of personal information may weaken adoption intention and perceived usefulness of digital retail technologies [20,35]. The qualitative findings further reinforced this moderation effect, as several interview participants stated that they preferred VTO platforms that clearly explained

how their data would be stored, protected, and used.

The integrated quantitative and qualitative analysis presented in Table 9 provides additional depth to the proposed conceptual framework by demonstrating how technological usefulness, behavioural confidence, sustainability orientation, and privacy evaluation interact simultaneously in shaping online shopping behaviour. The integrated findings indicate that VTO technology successfully bridges the gap between consumer expectations and actual product experiences by reducing uncertainty associated with online apparel shopping. Consumers using VTO systems reported higher confidence in purchase decisions, lower return frequency, and more sustainable consumption behaviour. However, the findings also demonstrate that technological effectiveness alone is insufficient for long-term adoption unless accompanied by transparent and ethical data privacy practices.

Overall, the findings establish VTO technology not only as a technological innovation but also as a sustainability-enabling mechanism within online fashion retailing. The study significantly extends existing literature by empirically integrating the Technology Acceptance Model and Privacy Calculus Theory to explain how consumers simultaneously evaluate technological usefulness, behavioural confidence, environmental outcomes, and privacy-related risks when adopting VTO systems. The results therefore suggest that the long-term success of VTO technology depends on both technological effectiveness and consumer trust toward data protection practices.

6. Conclusion

The rapid growth of online fashion retailing has increased concerns regarding high product return rates, reverse logistics inefficiencies, and environmental waste. In this context, the present study examined how VTO technology contributes to sustainable online fashion retailing by improving product selection accuracy and reducing return-related waste while considering the moderating role of data privacy concerns. By integrating the Technology Acceptance Model and Privacy Calculus Theory, the study provides a comprehensive explanation of how technological usefulness, behavioural confidence, sustainability outcomes, and privacy perceptions collectively influence consumer adoption of VTO technologies.

The findings revealed that VTO technology significantly improves consumers' ability to evaluate product fit, size, and appearance, thereby enhancing Accuracy of Product Selection and reducing uncertainty during online shopping. The structural model confirmed that VTO positively influences both Accuracy of Product Selection and Return Waste Reduction, while Accuracy of Product Selection itself significantly contributes to reducing unnecessary product returns. Furthermore, the mediation analysis demonstrated that VTO contributes to sustainability outcomes primarily through improved consumer decision-making accuracy, thereby establishing product selection accuracy as an important behavioural mechanism linking digital retail technologies with sustainable consumption outcomes.

The study also confirmed the moderating role of Data Privacy Concerns, indicating that consumers evaluate VTO technologies by balancing perceived shopping benefits against concerns related to biometric data collection and personal information sharing. While positive privacy perceptions strengthened trust and technology adoption, privacy concerns weakened consumer engagement despite the recognised usefulness of VTO systems. The qualitative findings further supported the quantitative results by revealing that VTO users experienced greater purchase confidence, lower return frequency, and more satisfactory shopping experiences, whereas frequent returners continued to face issues related to sizing inconsistency and product mismatch.

Overall, the study demonstrates that VTO technology functions not only as a customer engagement tool but also as a sustainability-enabling mechanism capable of reducing environmental waste associated with online fashion retailing. The findings suggest that the long-term success of VTO technologies will depend not only on technological effectiveness,

but also on retailers' ability to ensure transparency, consumer trust, and responsible data privacy practices within digital shopping environments.

6.1. Implications to theory and practice

6.1.1. Implication to theory

The results are a contribution to theory because the study combines both Technology Acceptance Model (TAM) and Privacy Calculus Theory as a way of explaining consumer adoption of VTO in online fashion retail. The results have shown that the perception of usefulness, ease of use, and privacy trust are very powerful determinants of adoption and have direct impact on product selection accuracy, purchase satisfaction, and return behaviour. Moreover, this paper illustrates that product selection accuracy is a mediating factor that minimizes the rate of returns and privacy concerns are moderating factors that expand on the existing models of digital retail technology adoption. Furthermore, the study supports the importance of previous studies that note the importance of AI-driven VTO and AR/VR technologies in the development of immersive experiences among consumers. Such technologies enhance not only the visualization and decision-making but also emotional involvement and intention to purchase. The integration of cognitive and affective aspects of consumer engagement and VTO makes this research valuable in providing a subtle insight into technology mediated online shopping behaviour.

6.1.2. Practical implications

In a practical sense, online fashion retailers can use VTO to enhance customer satisfaction, decrease returns and promote sustainable consumption. The retailers are advised to give the right visualization of the products, user-friendly user interfaces, and explicit privacy policies to build trust and adoption. The use of VTO in education, with an emphasis on such characteristics as the simulation of the size and styling recommendations, can improve consumer trust and decrease the level of cognitive load during online shopping. Furthermore, the application of AI-based personalization and recommendation system alongside the use of VTO have the potential to increase consumer engagement and intended purchase, providing competitive edge within an e-commerce platform. The retailers are in a position to track usage trends and feedback to keep on optimizing VTO features thus keeping up with consumer expectation and dynamic trends in technology.

6.1.3. Societal implications

VTO technology in online fashion retail is beneficial to the society in that it will decrease the product returns and, consequently, decrease the destruction of textiles and the environmental harm. VTO helps to develop responsible shopping behaviour by ensuring a better selection of products and allowing informed purchase choices, which will lead to a decrease in overconsumption and help to maintain more sustainable consumption patterns. Moreover, the increased privacy and data protection in VTO platforms should be a priority that will boost consumer trust and help to create safer and more ethical online shopping experiences. VTO technology enhances the digital shopping experience by ensuring proper visualization, sustainability, and digital trust, but also contributes to societal objectives of environmental preservation, ethical business and sustainable consumer behaviour.

6.1.4. Academic implications

The present study contributes to the academic literature by extending research on VTO technologies beyond consumer experience and purchase intention toward sustainability-oriented outcomes in online fashion retailing. Unlike prior studies that primarily focused on technology adoption or purchase intention, the present research extends existing literature by empirically connecting VTO adoption with sustainability outcomes through the mediating role of Accuracy of Product Selection and the moderating role of Data Privacy Concerns. The study

also contributes methodologically by adopting an explanatory sequential mixed-methods design that combines structural model analysis with qualitative consumer insights, thereby offering both statistical and contextual understanding of consumer behaviour. Furthermore, the findings contribute to the emerging discourse on how digital technologies can support environmentally sustainable consumption practices in e-commerce environments.

6.2. Limitations of the research

The research focused only on online fashion customers in Southern Karnataka, so the results may not fully apply to consumers in other regions, cultures, or product categories. The qualitative part of the study was based on a small number of interviews, and survey responses may also be influenced by social desirability or recall bias. In addition, the study mainly considered young adults, so the views of older consumers may not be fully represented. Some participants may also have had limited experience with VTO technology, which could affect their opinions. Finally, the study was conducted within a limited time period, so it may not capture changes in consumer behaviour or technology adoption over time.

6.3. Future research recommendations

Future studies may extend the geographic boundary to a national or cross-culture comparison and examine various consumer demographics and product type to enhance the generalizability. Research can also examine the impact of VTO adoption on purchase satisfaction, a return behaviour, and brand loyalty in the long run, combining the latest technologies like AI-controlled personalization, AR, VR, and social commerce to make the consumer engagement more engaging. Also, analysis of the environmental and sustainability effects of VTO-based return cuts, as well as privacy and ethical issues, can be used to create responsible and efficient digital retail solutions.

Ethical statement

The study was done with strict adherence to the ethical standards. The respondents volunteered to participate, and informed consent was received by all of them. The research participants were made aware on the intent of the research, the intended use, and their entitlement to drop out any time. Personally identifiable information was not gathered and all the responses were treated with their privacy. The Institutional Ethics Committee of YIASCAM gave ethical approval to conduct the research under the Reference Number YIASCAM/ST/SRB/01/MGT/46/2025. The data were utilized in an academic purpose. Particular attention was paid

to the consideration of privacy-related questions to provide comfort and trust to the participants.

Informed consent

All participants were informed and gave their respective consents. All participants gave their free and voluntary consent, free from any form of coercion. During data collection, the participants were aware of the objectives of the study and any risks or benefits they might incur, and thus were able to provide their consent in writing. Participant autonomy and confidentiality were therefore ensured in accordance with ethical standards.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors of this manuscript used Grammarly and ChatGPT (OpenAI) for language improvement, grammar correctness, and sentence structure enhancement in developing this document. All content generated with the aid of Grammarly and ChatGPT was thoroughly examined, edited, and approved by the authors; so, they assume complete responsibility for any content that appears in the published manuscript.

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Shifa Sheikh: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **S.M. Riha Parvin:** Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft. **Safa Sayeed:** Investigation, Methodology, Supervision, Writing – review & editing. **Niyaz Panakaje:** Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Questionnaire

Impact of virtual try-on technology on product selection accuracy and return waste reduction in online fashion shopping: assessing the role of data privacy concerns

This survey is designed for the purpose of conducting a study on "Impact of Virtual Try-On Technology on Product Selection Accuracy and Return Waste Reduction in Online Fashion Shopping: Assessing the Role of Data Privacy Concerns" as a part of academic research. The main intention of this study is to understand how Virtual Try-On (VTO) technology influences online shopping experiences, improves product selection accuracy, reduces unnecessary product returns, and supports environmentally responsible consumption behaviour in online fashion retailing. The study also aims to examine how data privacy concerns influence consumers' willingness to adopt and use Virtual Try-On technologies during online shopping.

Your participation in responding to this questionnaire is voluntary; however, your valuable participation is highly appreciated and will significantly contribute to research in the area of digital retail technology, sustainable online shopping, and consumer behaviour. Therefore, I kindly request you to spare your valuable 10 minutes to provide accurate and experience-based responses for this academic study.

All information collected in this study will be strictly kept confidential and will be used only for the purpose of the above-mentioned study.

Section A - Demographic details

Sl. No	Variables				
1.	Name (optional):				
2.	Age	18-25	26-33	33-40	41 and above
3.	Gender	Male	Female		
4.	Education Level	High School	PUC	Undergraduate	
		Postgraduate	Doctorate	Diploma	
5.	Occupation	Student	Working Professional		Self-Employed
		Homemaker	Unemployed		
6.	Monthly Income	Below Rs. 10,000		Rs. 10,000 - Rs. 25,000	
		Rs.25001 - Rs. 50,000		Rs. 50,000 and above	
7.	How frequently do you shop online?	Rarely (less than once a month)		Occasionally (1-2 times a month)	
		Frequently (3-5 times a month)		Very Frequently (more than 5 times a month)	
8.	How much do you spend on online shopping?	Less than Rs. 1,000		Rs. 1,000 - Rs. 5000	
		Rs. 5001 - Rs. 10000		Rs. 10,001 and above	
9.	How frequently do you use Virtual try on feature when buying fashion apparel online?	Rarely	Occasionally	Frequently	Very Frequently
10.	On which platforms have you used Virtual Try On?	Myntra	Amazon	Ajio	Tata Cliq
		Nykaa Fashion	Other:		
11.	What are your concerns over shopping online when using VTO?	Virtual fit is not accurate	Product does not look the same in real	Privacy concerns	VTO is not available for all products
		No concerns			
12.	How often have you returned a fashion apparel bought online by using VTO?	Rarely	Occasionally	Frequently	Never returned
13.	My willingness to share personal data for accurate product selection.	Very willing	Somewhat willing	Neutral	Unwilling
		Strongly unwilling			

Section B – Impact of VTO

II. Kindly indicate your agreement level on the following statement (Kindly tick (✓) your opinion) (5– Strongly Agree; 4–Agree; 3–Neutral; 2–Disagree; 1–Strongly Disagree)

Sl. No.	Virtual try on	Rating				
		5	4	3	2	1
1.	I find VTO tools useful for making accurate shopping choices online.					
2.	I understand how to use VTO without much effort.					
3.	VTO makes me feel confident about my purchasing decisions.					
4.	I would recommend VTO to others for online shopping.					
5.	VTO accurately represents how a product would look on me.					
	Perceived usefulness					
1.	Using the Virtual Try-On (VTO) feature would help me make better apparel purchase decisions.					
2.	The VTO feature would make online shopping for clothes more convenient for me.					
3.	Using VTO would improve the accuracy of my size selection.					
4.	The VTO feature would reduce the chances of returning products due to wrong fit.					
5.	Overall, I find the VTO feature useful for improving my online shopping experience.					
	Adoption intention					
1.	I intend to use the VTO feature when shopping for apparel online.					
2.	I am willing to try the VTO feature the next time I shop online.					
3.	I will continue using VTO if it becomes available on fashion websites I visit.					
4.	I would recommend using the VTO feature to others who shop for apparel online.					
5.	I am likely to rely on the VTO feature before making my final purchase decision.					

III. Kindly indicate your agreement level on the following statement (Kindly tick (✓) your opinion) (5– Strongly Agree; 4–Agree; 3–Neutral; 2–Disagree; 1–Strongly Disagree)

Sl. No.	Return waste reduction	Rating				
		5	4	3	2	1
1.	Using VTO helps me reduce the number of items I return from online purchases.					
2.	VTO reduces shipping waste caused by product returns.					
3.	I believe VTO is an effective tool for making online shopping more environmental friendly.					
4.	I prefer to shop from online retailers that implement technologies like VTO to reduce environmental impact.					
5.	Using VTO aligns with my personal values regarding environmental responsibility in consumption.					

IV. Kindly indicate your agreement level on the following statement (Kindly tick (✓) your opinion) (5– Strongly Agree; 4–Agree; 3–Neutral; 2–Disagree; 1–Strongly Disagree)

Sl. No.	Accuracy of product selection	Rating				
		5	4	3	2	1
1.	VTO helps me feel more confident about selecting the correct product size					
2.	The virtual preview provided by VTO accurately matches the appearance of the real product.					
3.	Using VTO reduces the chance of choosing the wrong size or fit.					
4.	VTO improves my ability to assess how a style will look on me.					
5.	I feel confident that my product choice is accurate after using VTO					

V. Kindly indicate your agreement level on the following statement (Kindly tick (✓) your opinion) (5– Strongly Agree; 4–Agree; 3–Neutral; 2–Disagree; 1–Strongly Disagree)

Sl. No.	Data privacy concern	Rating				
		5	4	3	2	1
1.	I am uncomfortable providing my body measurements for VTO.					
2.	I am concerned about the possibility of my personal information being accessed by third parties while using VTO.					
3.	My willingness to use virtual try-on feature reduces if I have to share sensitive personal data.					
4.	I worry that my personal data collected could be used for other purposes as well.					

VI. Kindly indicate your agreement level on the following statement (Kindly tick (✓) your opinion) (5– Strongly Agree; 4–Agree; 3–Neutral; 2–Disagree; 1–Strongly Disagree)

Sl. No.	Data privacy concern	Rating				
		5	4	3	2	1
1.	I am aware that my biometric data is collected when using VTO.					
2.	I trust that VTO platforms handle my personal data responsibly.					
3.	I believe VTO platforms are transparent about their data privacy policies.					
4.	I am willing to share my personal information to use VTO features.					

-Thank you for your Valuable Response....

Data availability

Data will be made available on request.

References

[1] Bain & Company, & F., *How India shops online 2023*, Bain & Company, 2023.

[2] DFU Publications. (2025). *The high cost of growth: How returns are reshaping fashion e-commerce in India*.

[3] Frei R, Jack L, & Brown S. (2020). *Product returns: a growing problem for business, society and environment* Int. J. Oper. Prod. Manage.

[4] IndiaSpend, *Textile waste in India: Environmental impact of rising apparel consumption and returns*, IndiaSpend (2024).

[5] S. Sanalkumar, Consumer product returns post online purchases: systematic research, Int. J. Manage. (IJM) 11 (9) (2020) 171–186, <https://doi.org/10.34218/IJM.11.9.2020.018>.

[6] *The Economic Times*, *More than a third of fashion and footwear products get returned in online shopping*, *The Economic Times*, 2026.

[7] M. Ali, S. Siraj, M. Ahmad, Sustainable fashion through the lens of AR/VR: conceptual framework, Int. J. Retail Distrib. Manage. 54 (1) (2026) 77–92, <https://doi.org/10.1108/IJRDM-09-2024-0506>.

[8] L.S. Vitola, *Digital Innovation for Sustainable Fashion: Virtual Fitting Rooms as a Strategy to Minimize Waste and E-commerce Returns*, Toronto Metropolitan University, 2024.

[9] C.-C. Luan, S. Zhang, W.-L. Chen, Augmented reality try-ons as phygital integration technology for sustainable fashion products, Int. J. Retail Distrib. Manage. 54 (3) (2026) 344–359, <https://doi.org/10.1108/IJRDM-03-2025-0215>.

[10] C. Bozzi, M. Neves, C. Mont'alvão, Fashion E-tail and the impact of returns: mapping processes and the consumer journey towards more sustainable practices, Sustainability (Switzerland) 14 (9) (2022) 5328, <https://doi.org/10.3390/su14095328>.

[11] A. Nestler, N. Karessli, K. Hajjar, R. Weffer, R. Shirvany, SizeFlags: reducing size and fit related returns in fashion e-commerce, in: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2021, pp. 3432–3440, <https://doi.org/10.1145/3447548.3467160>.

[12] T. Rintamäki, M.T. Spence, H. Saarijärvi, J. Joensuu, M. Yrjölä, Customers' perceptions of returning items purchased online: planned versus unplanned product returns, Int. J. Phys. Distrib. Logistics Manage. 51 (4) (2021) 403–422, <https://doi.org/10.1108/IJPDLM-10-2019-0302>.

[13] A.K. Seewald, T. Wernbacher, A. Pfeiffer, N. Denk, M. Platzer, M. Berger, T. Winter, Towards minimizing e-commerce returns for clothing, in: *ICAART 2019 - Proceedings of the 11th International Conference on Agents and Artificial Intelligence 2*, 2019, pp. 801–808, <https://doi.org/10.5220/0007568508010808>.

[14] M. Zoubi, H. Estaitia, A. Morshed, L.T. Khrais, E. Haikal, M. AlSheikh, Augmented reality and sustainable luxury: transforming fashion retail in the UAE, Technol. Sustainability 4 (3) (2025) 275–292, <https://doi.org/10.1108/TECHS-03-2025-0068>.

[15] Fele, B., Lampe, A., Peer, P., & Struc, V. (2023). *C-VTON: context-driven image-based virtual try-on network*.

[16] D. Marelli, S. Bianco, G. Ciocca, Designing an AI-based virtual try-on web application, Sensors 22 (10) (2022) 3832, <https://doi.org/10.3390/s22103832>.

[17] D. Song, X. Zhang, J. Zhou, W. Nie, R. Tong, M. Kankanhalli, A.-A. Liu, Image-based virtual try-on: a survey, <https://arxiv.org/abs/2311.04811>, 2024.

[18] Neuberger, A., Borenstein Bar, E., Oks, H. E., & Alpert, S. (2020). *Image based virtual try-on network from unpaired data*.

[19] Y. Gao, J. Liang, The impact of AI-powered try-on technology on online consumers' impulsive buying intention: the moderating role of brand trust, Sustainability (Switzerland) 17 (7) (2025) 2789, <https://doi.org/10.3390/su17072789>.

[20] S.yi Youn, J. Hwang, L. Zhao, J.B Kim, Privacy paradox in 3D body scanning technology: the effect of 3D virtual try-on experience in the relationship between privacy concerns and mobile app adoption intention, Humanities Social Sci. Commun. 10 (1) (2023), <https://doi.org/10.1057/s41599-023-01632-y>.

[21] T. Zhang, W.Y.C. Wang, L. Cao, Y. Wang, The role of virtual try-on technology in online purchase decision from consumers' aspect, Internet Res. 29 (3) (2019) 529–551, <https://doi.org/10.1108/IntR-12-2017-0540>.

[22] M. Niederlaender, A.N. Lodi, S. Gry, R. Biswas, D. Werth, Garment returns prediction for ai-based processing and waste reduction in e-commerce, Int. Conf.

- Agents Artif. Intell. 2 (2024) 156–164, <https://doi.org/10.5220/0012321300003636>.
- [23] J.Y. Jang, Analyzing visual behavior of consumers in a virtual reality fashion store using eye tracking, *Fashion Textiles* 10 (1) (2023), <https://doi.org/10.1186/s40691-023-00345-9>.
- [24] E. Sardanou, M. Briana, Platform-driven sustainability in e-commerce: consumer behavior toward recycled fashion, *Recycling* 10 (4) (2025) 161, <https://doi.org/10.3390/recycling10040161>.
- [25] A.R. Asadi, Y. Zhang, H. Said, What do personas say about privacy & security: a systematic literature review through human-AI collaboration, *J. Ambient. Intell. Humaniz. Comput.* 16 (11–12) (2025) 1175–1190, <https://doi.org/10.1007/s12652-025-05007-w>.
- [26] A. Ivanov, Y. Mou, L. Tawira, Avatar personalisation vs. privacy in a virtual try-on app for apparel shopping, *Int. J. Fashion Des. Technol. Educ.* 16 (1) (2023) 100–109, <https://doi.org/10.1080/17543266.2022.2125589>.
- [27] B.A. Ojokoh, F.O. Isinkaye, M. Zhang, J.J. Tom, A.J. Gabriel, O. Afolabi, B. Afolabi, Privacy and security in recommenders: an analytical review, *Artif. Intell. Rev.* 58 (11) (2025), <https://doi.org/10.1007/s10462-025-11333-4>.
- [28] A. Ragab, M.M. Id, A. Youssef, Try on, Spied on? : Privacy Analysis of Virtual Try-On Websites and Android Apps, 2023. <https://github.com/virtualtryon2023/open-wpm-cookies-and-scripts-extension>.
- [29] I. Mutambik, J. Lee, A. Almuqrin, J.Z. Zhang, Transitioning to smart cities in gulf cooperation council countries: the role of leadership and organisational culture, *Sustainability (Switzerland)* 15 (13) (2023) 10490, <https://doi.org/10.3390/su151310490>.
- [30] *The Indian Express*, Online retailers grapple with 30% product returns, cancellations during festive season, 2018.
- [31] A. Merle, S. Senecal, A. St-Onge, Whether and how virtual try-on influences consumer responses to an apparel web site, *Int. J. Electronic Commerce* 16 (3) (2012) 41–64, <https://doi.org/10.2753/JEC1086-4415160302>.
- [32] K. Sekri, O. Bouzaabia, H. Rzem, D. Juárez-Varón, Effects of virtual try-on technology as an innovative e-commerce tool on consumers' online purchase intentions, *Eur. J. Innov. Manage.* 28 (2025) 4041–4060, <https://doi.org/10.1108/EJIM-05-2024-0516>.
- [33] C. Chen, J. Ni, P. Zhang, Virtual try-on systems in fashion consumption: a systematic review, in: *Applied Sciences (Switzerland)*, 14, Multidisciplinary Digital Publishing Institute (MDPI), 2024, <https://doi.org/10.3390/app142411839>.
- [34] R. Shiba, A. Fadel, A. Awad, K. Emad, R.El Gohary, *An Immersive Virtual Try-On System with Augmented Reality*, Vol. 2, 2025.
- [35] I. Mutambik, J. Lee, A. Almuqrin, J.Z. Zhang, A. Homadi, The growth of social commerce: how it is affected by users' privacy concerns, *J. Theoret. Appl. Electronic Commerce Res.* 18 (1) (2023) 725–743, <https://doi.org/10.3390/jtaer18010037>.
- [36] S. Barta, R. Gurra, C. Flavián, How Augmented reality increases engagement through its impact on risk and the decision process, *Cyberpsychol. Behav. Social Netw.* 26 (3) (2023) 177–187, <https://doi.org/10.1089/cyber.2022.0087>.
- [37] H. Hwangbo, E.H. Kim, S.H. Lee, Y.J. Jang, Effects of 3D virtual "try-on" on online sales and customers' purchasing experiences, *IEE Access*. 8 (2020) 189479–189489, <https://doi.org/10.1109/ACCESS.2020.3023040>.
- [38] V. Lavoye, J. Sipilä, J. Mero, A. Tarkiainen, The emperor's new clothes: self-exploratory engagement in virtual try-on service experiences positively impacts brand outcomes, *J. Serv. Market.* 37 (10) (2023) 1–21, <https://doi.org/10.1108/JSM-04-2022-0137>.
- [39] Tandon, U. Ertz, M., & Sakshi. K. (2021). *POD mode of payment, return policies and virtual-try-on technology as predictors of trust: an emerging economy case*.
- [40] F.D. Davis, Perceived usefulness, perceived ease of use, and user acceptance of information technology, *MIS Quarterly* 13 (3) (1989) 319–340, <https://doi.org/10.2307/249008>.
- [41] Culnan and Armstrong, Information privacy concerns, procedural fairness, and impersonal trust: an empirical investigation on JSTOR. <https://www.jstor.org/stable/2640390>, 1999.
- [42] H.H.Y. Chan, C. Henninger, R. Boardman, M. Blazquez Cano, The adoption of digital fashion as an end product: a systematic literature review of research foci and future research agenda, *J. Global Fashion Market.* 15 (1) (2024) 155–180, <https://doi.org/10.1080/20932685.2023.2251033>.
- [43] Gouveia De Almeida, M. I. (2021). *Consumers acceptance of artificial intelligence virtual try-on systems when shopping apparel online*.
- [44] F.D. Davis, R.P. Bagozzi, P.R. Warshaw, User acceptance of computer technology: a comparison of two theoretical models, *Manage. Sci.* 35 (8) (1989) 982–1003, <https://doi.org/10.1287/mnsc.35.8.982>.
- [45] Costa, A. Marozzo, V., & Abbate, T. (2025). *Consumers' attitudes toward virtual try-on technology: an extended TAM model*. 53(13), 184.
- [46] V. Lavoye, J. Mero, A. Tarkiainen, Consumer behavior with augmented reality in retail: a review and research agenda, *Int. Rev. Retail Distrib. Consumer Res.* 31 (3) (2021) 299–329, <https://doi.org/10.1080/09593969.2021.1901765>.
- [47] S. Gupta, R. Kiran, R.K. Sharma, Analyzing the complexities of chatbots, virtual try-on and digital payment mode on customer satisfaction: an empirical study of online shoppers, *Eur. J. Manage. Bus. Econ.* (2025) 1–18, <https://doi.org/10.1108/EJMBE-08-2023-0260>.
- [48] V.W. Anelli, L. Belli, Y. Deldjoo, T. Di Noia, A. Ferrara, F. Narducci, C. Pomo, Pursuing privacy in recommender systems: The view of users and researchers from regulations to applications, in: *RecSys 2021 - 15th ACM Conference on Recommender Systems*, 2021, pp. 838–841, <https://doi.org/10.1145/3460231.3473326>.
- [49] Y.F. Wu, E.Y. Kim, Users' perceptions of technological features in augmented reality (AR) and virtual reality (VR) in fashion retailing: a qualitative content analysis, *Mobile Inf. Syst.* 2022 (2022) 1–13, <https://doi.org/10.1155/2022/3080280>.
- [50] *Textile Waste A Growing Environmental Concern*, IndiaSpend. <https://www.indiaspend.com/industry/textile-waste-a-growing-environmental-concern-932145>, 2024.
- [51] *The Economic Times*, ecommerce: More than a third of fashion and footwear products get returned in online shopping: Report - The Economic Times, 2024. <https://economictimes.indiatimes.com/industry/cons-products/fashion/-cosmetics/-jewellery/more-than-a-third-of-fashion-and-footwear-products-get-returned-in-online-shopping-report/articleshow/112142214.cms?from=mdr>.
- [52] T. Islam, A. Miron, X. Liu, Y. Li, Deep learning in virtual try-on: a comprehensive survey, *IEE Access* 12 (2024) 29475–29502, <https://doi.org/10.1109/ACCESS.2024.3368612>.
- [53] K. Niinimäki, L. Hassi, Emerging design strategies in sustainable production and consumption of textiles and clothing, *J. Clean. Prod.* 19 (16) (2011) 1876–1883, <https://doi.org/10.1016/j.jclepro.2011.04.020>.
- [54] P. Battistoni, M. Di Gregorio, M. Romano, M. Sebillio, G. Vitiello, A. Brancaccio, Interaction design patterns for augmented reality fitting rooms, *Sensors* 22 (3) (2022) 982, <https://doi.org/10.3390/s22030982>.
- [55] B. Gündür, D. Deniz, Impacts of augmented reality marketing on gen-z fashion consumers' behavioral intention, *Tekstil ve Muhendis* 31 (136) (2024) 253–268, <https://doi.org/10.7216/tekstmu.1529591>.
- [56] X. Li, Privacy-Preserving Feature Attribution Explanations for Large-Scale Recommendation Systems: A Differential Privacy Approach, *Journal of Science, Innovation & Social Impact EISSN: PISSN (Number 1)*, 2025.
- [57] M.E. Ioannou, *Perceptual Maps: Methodology, Sample Characteristics and General Data Analysis of Customers-Perception on Virtual Try-on Features* (Master's thesis, Universidade NOVA de Lisboa (Portugal)), 2023.
- [58] I. González Romero, Á. Ortiz Bas, J.C. Prado Prado, The perspective of key stakeholders on the sustainable design of reverse logistics in e-commerce: an interpretive sensemaking review, *Int. J. Prod. Manage. Eng.* 13 (1) (2025) 122–136, <https://doi.org/10.4995/IJPMME.2025.22508>.
- [59] T. Issenhu, J. Mary, C. Calauzènes, Do not mask what you do not need to mask: a parser-free virtual try-on. <http://arxiv.org/abs/2007.02721>, 2020.
- [60] A. Awad, Augmented reality as a tool for customer engagement: impact on brand perception in retail, *TEM J.* (2025) 2111–2123, <https://doi.org/10.18421/tem143-18>.
- [61] S.H.A. Kazmi, R.R. Ahmed, K.A. Soomro, A.R. Hashem E, H. Akhtar, V. Parmar, Role of augmented reality in changing consumer behavior and decision making: case of Pakistan, *Sustainability (Switzerland)* 13 (24) (2021) 14064, <https://doi.org/10.3390/su132414064>.
- [62] Y. Zheng, T. Zhang, J. Bao, D. Chen, M. Zeng, High-fidelity instructional fashion image editing, *Graph Models* 135 (2024) 101223, <https://doi.org/10.1016/j.gmod.2024.101223>.
- [63] F.T. Dezaki, H. Arora, R. Suresh, A. Banitalebi-Dehkordi, Automated material properties extraction for enhanced beauty product discovery and makeup virtual try-on. <http://arxiv.org/abs/2312.00766>, 2023.
- [64] Hadi. K., & Avagyan. V. (2021). *Say Hello to the Future of Beauty*.
- [65] L. Tawira, A. Ivanov, Leveraging personalization and customization affordances of virtual try-on apps for a new model in apparel m-shopping, *Asia Pacific J. Market. Logistics* 35 (2) (2023) 451–471, <https://doi.org/10.1108/APJML-09-2021-0652>.
- [66] Y. Liu, Y. Liu, S. Xu, K. Cheng, S. Masuko, J. Tanaka, Comparing vr-and ar-based try-on systems using personalized avatars, *Electronics (Switzerland)* 9 (11) (2020) 1814, <https://doi.org/10.3390/electronics9111814>.
- [67] M.S. Akter, S.A. Khan, N. Sultana, M.T. Islam, Unpacking how virtual try-on technology influences consumer decision-making in online shopping: insights from Bangladesh, *Hum. Behav. Emerg. Technol.* 2025 (1) (2025), <https://doi.org/10.1155/hbe2/7070627>.
- [68] C. Guo, X. Zhang, The impact of AR online shopping experience on customer purchase intention: An empirical study based on the TAM model, *PLoS. One* 19 (8 August) (2024) e0309468, <https://doi.org/10.1371/journal.pone.0309468>.
- [69] J.W. Creswell, J.D. Creswell, *Diseño de investigación. Enfoques cualitativos, cuantitativos y mixtos. Book*, 2017, p. 304. https://books.google.com/books/abou/Research_Design.html?id=335ZdwAAQBAJ.
- [70] S.L. Siedlecki, Understanding descriptive research designs and methods, *Clin. Nurse Specialist* 34 (1) (2020) 8–12, <https://doi.org/10.1097/NUR.0000000000000493>.
- [71] J.F. Hair, M.C. Howard, C. Nitzl, Assessing measurement model quality in PLS-SEM using confirmatory composite analysis, *J. Bus. Res.* 109 (2020) 101–110, <https://doi.org/10.1016/j.jbusres.2019.11.069>.
- [72] J.W. Creswell, *Research design : qualitative, quantitative, and mixed methods approaches*, Sage Publications, 2003.
- [73] Hatzidaki, M. (2023). Market segmentation & expert interviews: a methodological approach to researching consumer behaviour towards virtual try-ons in the online fashion retail industry.
- [74] J.F. Hair, J.J. Risher, M. Sarstedt, C.M. Ringle, When to use and how to report the results of PLS-SEM, *Eur. Bus. Rev.* 31 (1) (2019) 2–24, <https://doi.org/10.1108/EBR-11-2018-0203>.
- [75] E.B. Diop, S. Zhao, T. Van Duy, An extension of the technology acceptance model for understanding travelers' adoption of variable message signs, *PLoS. One* 14 (4) (2019) e0216007, <https://doi.org/10.1371/JOURNAL.PONE.0216007>.
- [76] Y. Deng, H. Shen, X. Ji, Exploring virtual fashion consumption through the emotional three-level theory: reflections on sustainable consumer behavior, *Sustainability (Switzerland)* 16 (13) (2024) 5818, <https://doi.org/10.3390/SU16135818>.